REDD+ Projects Work: Evidence from Afro-Colombian Communities

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Abstract

Global efforts to mitigate climate change include Reducing Emissions from Deforestation and Forest Degradation (REDD+). However, evidence on its effectiveness remains mixed, with many studies finding limited impacts on deforestation and overlooking potential effects on other lan use changes. This study evaluates the effectiveness of collective REDD+ agreements signed by Afro-Colombian communities in curbing both deforestation and coca cultivation. Using satellite data at 30 m × 30 m resolution for forest cover (900 m^2 per pixel) and 1 km × 1 km resolution for coca crops (1 km^2 per pixel), we find that REDD+ projects significantly reduced deforestation by an average of 33 m^2 per pixel and coca cultivation by 0.33 ha / km2, which is approximately 9,695 hectares of avoided deforestation and 7,098 hectares of contained coca crop expansion annually. These significant impacts, which stand in contrast to findings from previous REDD+ evaluations, likely stem from the way these collective agreements strengthen local monitoring systems, enhance social cohesion, and generate expectation of improved economic conditions.

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1 Introduction

Deforestation and forest burning are major drivers of climate change due to their substantial contribution to global greenhouse gas emissions. Forests serve as carbon sinks, absorbing CO_2 from the atmosphere and storing it in plant biomass and soil (Intergovernmental Panel On Climate Change, 2023). The tropical regions, home to the most biodiverse and carbon dense forests on the planet (Artaxo et al., 2022; Brienen et al., 2015; Tracewski et al., 2016), are experiencing high rates of deforestation, especially in Colombia. In response to the increasing loss of forests, particularly in tropical areas, market-based approaches have been established to encourage ecosystem conservation by local communities, ensuring sustained provision of ecosystem services and therefore contributing to the mitigation of climate change (Angelsen, 2008; Angelsen, 2014; Engel, 2016; Engel et al., 2008).

A prominent market-based conservation tool is Payments for Ecosystem Services (PES), which offers conditional incentives to landholders or communities for verifiable actions that supply ecological benefits (Engel et al., 2008). REDD+ adopts the same logic in its "results" (third) phase, linking rewards to documented reductions in forest-based carbon emissions (Angelsen, 2008; Angelsen, 2014; Naime et al., 2022). Yet REDD+ differs from conventional PES in both scope and funding: its overriding goal is climate mitigation through carbon sequestration, and—although most countries still finance REDD+ largely with public, results-based aid—Colombia stands out for channeling a substantial share of REDD+ finance through voluntary carbon-credit markets, encouraged by its carbon-tax offset scheme.

The voluntary carbon market has experienced exponential growth, with the value of carbon credit transactions increasing from approximately $530 million in 2012 to $1.9 billion in 2022(F. T. E. Marketplace, 2023). Nature-based credits, especially those from REDD+, have become a significant part of the market, making up 74% of such credits issued by 2022 (Probst et al., 2023). Colombia has played a notable role in these projects, with its highest participation occurring in 2018. During that year, Colombia was responsible for 48% of the credits from forestry and related activities traded on the international market. [1]

Despite this growth trajectory the performance and credibility of the REDD+ within the voluntary carbon market face significant scrutiny. In consequence, after the broader market peaked in 2021, it experienced a contraction with a 56% drop in transacted volume from 2021 to 2023 (E. Marketplace, 2024). This economic decline occurred in the middle of an escalating credibility crisis, driven by ongoing doubts about the additionality of projects, specifically whether they truly prevent emissions that would not have occurred otherwise (West et al., 2020, 2023).

There is still no consensus regarding the impact of REDD+ on deforestation and degradation, and even less is known about its performance in collective land-tenure contexts. On one hand, some studies have identified a positive and significant impact on curbing deforestation

[1] While in 2018, 48% of AFOLU (Agriculture, Forestry, and Other Land Use) carbon offsets originated from Colombia, by 2022, this proportion had dropped to only 5% (Haya et al., 2023). This decline can be attributed in part to the fact that a significant number of projects have not yet completed the certification processes required to issue carbon credits post-2018. This observation is supported by data from the Voluntary Registry Offsets Database, developed by the Berkeley Carbon Trading Project, which compiles information on all carbon offset projects, credit issuances, and retirements listed worldwide by four major voluntary offset project registries (Haya et al., 2023)

(Guizar-Coutiño et al., 2022; Simonet et al., 2019). On the other hand, others have highlighted REDD+'s limitations in addressing key deforestation drivers, raising doubts about additionality and the long-term sustainability of reported gains (Badgley et al., 2022; West et al., 2020, 2023). Beyond concerns about additionality, issues of equity have also been questioned (Bayrak & Marafa, 2016).

Furthermore, research to date largely focuses on deforestation alone, overlooking broader land-use changes and potential unintended impacts on other licit or illicit activities. Although REDD+ does not explicitly aim to reduce illicit crops or condition its benefits on their presence, unlike alternative development projects, understanding its influence on these dynamics is essential. Illicit crops remain both a direct and an important indirect driver of land-use change (Dávalos et al., 2016; P. J. Murillo-Sandoval et al., 2023). Moreover, drug policies and interventions designed to curb illicit crop production can result on more deforestation (Londoño et al., 2024).

Finally, despite the fact that a substantial share of the world's tropical forests is communally owned, there is a notable lack of research on how REDD+ initiatives function within collective land-tenure systems. In the global South, 27% of forests—and 33% in Latin America—are under collective ownership (Agrawal, 2007; FAO & FILAC, 2021). In Colombia specifically, 36% of the land is collectively owned and 53% of the country's natural forests fall under collective titles held by ethnic communities (FAO, 2020). This prevalence of collective ownership makes Colombia a particularly relevant case for examining how community-based conservation and market based initiatives might help curb deforestation.

By analyzing the interaction between market based conservation and collective governance, this article estimates the causal impact of collective REDD+ on deforestation and its broader effects on coca crops[2] within Afro-Colombian collective lands. To achieve this, we leverage high-resolution, granular satellite data at the pixel level, constructing a detailed panel dataset covering 17 REDD+ projects implemented across 33 community councils, encompassing over 2 million hectares between 2000 and 2021. Our empirical strategy relies on a staggered difference-in-differences (DiD) methodology following Callaway and Sant'Anna (2021), this approach explicitly accounts for the fact that different community councils entered into REDD+ agreements at various points in time.

The results show that the REDD+ projects significantly contained deforestation pressures within the participating communities. We estimate an average reduction of 33 m^2 per pixel relative to control units, which translates to approximately 9,695 hectares of avoided deforestation annually. These initiatives also limited the expansion of coca cultivation, achieving an average reduction of 0.33 ha/km^2; this represents an estimated 7,098 hectares per year of contained growth.

The robustness of these central findings is confirmed through a series of sensitivity analyses, including the application of alternative robust DiD estimators (e.g., Sun and Abraham (2021) and Wooldridge (2021)), the use of different control groups, validation against an alternative global deforestation dataset (Hansen et al., 2013), and checks ensuring results are

[2]Throughout this paper, "*coca crops*" refers to plantings of the coca shrub (*Erythroxylum coca* and *E. novogranatense*). In Colombia the cultivation of coca is illegal, except for small plots expressly permitted for traditional, medicinal, or scientific uses (Law 30 of 1986; 1961 Single Convention on Narcotic Drugs). While a minor share of the harvested leaf is chewed, brewed, or otherwise used in ancestral cultural practices, the vast majority is diverted to the illicit production of cocaine.

not driven by the size of the councils. This article, therefore, offers compelling evidence that these collective REDD+ schemes, operating within specific governance structures, can effectively mitigate both deforestation and coca expansion, providing a counterpoint to studies reporting limited REDD+ impacts elsewhere.

We study four mechanisms that explain these outcomes: (1) governance and social capital, (2) individual and collective benefits, (3) the role of intermediaries, and (4) surveillance and global climate governance. The components of REDD + projects support aspects of social capital, which the literature has identified as pivotal for the governance of the commons. Notably, this includes the participatory construction of rules, the revision of internal boundaries and collective definition of land use, monitoring, and control mechanisms (Brownson et al., 2019; Hayes et al., 2019; Lobo & Vélez, 2022; Naime et al., 2022). Regarding economic incentives, despite the collective nature of compensation in such projects, individuals receive indirect monetary benefits since many projects promote, finance, or enhance the qualification of existing agricultural practices on previous existing agricultural land. Additionally, job creation related to forest and biodiversity patrol and monitoring emerges. Furthermore, the collective resources invested in common goods, such as infrastructure within the councils, impact the individuals. Lastly, the elements of global climate surveillance in which these projects are embedded, attracting a panoptic gaze through satellite imagery, may contribute to changes in practices and could be strongly associated with the findings related to coca crops.

Our primary contribution addresses the current issues surrounding the trustworthiness of carbon credits by offering strong causal evidence that REDD+ is indeed capable of achieving its promised environmental results. Beyond that, we break new ground in the evaluation of REDD+ by focusing on collective lands and their distinctive socio-environmental dynamics. First, this study represents the first comprehensive assessment of REDD+ programs implemented within collective lands. And to the best of our knowledge, it is one of the few studies and first in Colombia to infer the causal impact of REDD+. This is significant, as collective land presents unique challenges and dynamics distinct from those of individual or state-owned land, especially in terms of governance, communal decision-making, and monitoring.

Second, our research expands the evaluation of REDD+ programs beyond the conventional focus on deforestation and carbon sequestration to investigate its impact on illicit crops. This focus enables a nuanced understanding of the program's effects in regions characterized by complex environmental and social landscapes with interactions among licit and illicit economies.

Third, our work provides insights into the intersection between environmental governance and drug policy by assessing whether REDD+ can serve as a viable mechanism for mitigating the environmental harms associated with illicit economies. This study is the first to document the positive impact of an environmental policy on reducing coca cultivation, demonstrating how conservation incentives can influence land use decisions beyond their primary scope. These findings are particularly relevant given increasing concerns over the integrity of carbon markets, as buyers seek offsets that demonstrate verifiable co-benefits (F. T. E. Marketplace, 2023). Furthermore, they provide an alternative perspective to traditional drug eradication policies, which have not only failed to reduce coca crops but have also contributed to deforestation (DNP, 2024). In this context, the additional environmen-

tal benefits of controlling illicit crops through REDD+ could enhance the credibility and marketability of carbon credits, opening new opportunities for conservation finance.

The paper proceeds as follows. Section 2 analyses the relevant literature on PES and REDD+ as well as the role of illicit crops and drug policies as drivers of deforestation. Section 3 discusses the theory behind collective payments. Section 4 describes the context and data. Section 5 presents our identification strategy. Section 6 presents our main results on coca expansion and deforestation. Section 7 discusses potential mechanisms. Section 8 concludes.

2 Relevant literature: PES and REDD+ and illicit crops as drivers of deforestation

This section presents an overview of empirical evidence on the impact of PES and REDD+, analyzing their efficacy, challenges, and the opportunities they offer within market-based conservation approaches, particularly in collectively owned lands. It also discusses the literature exploring the role of illicit crops and drug policies as drivers of deforestation.

2.1 Empirical Evidence of PES and REDD+

Impact of PES in land use. There is a substantial corpus of impact studies examining the potential of Payment for Ecosystem Services (PES) schemes to provide additional environmental benefits, particularly in terms of controlling deforestation across various countries (for a review see: Wegner (2016)). Although PES programs have demonstrated a capacity to reduce deforestation rates, the extent of this effect is often moderate or marginal. Notable studies reporting positive, yet restrained impacts include those conducted in Costa Rica (Robalino & Pfaff, 2013), Mexico (Alix-Garcia et al., 2012, 2018; Charoud et al., 2023), Uganda (Jayachandran et al., 2017), Brazil (Oliveira Fiorini et al., 2020), and Cambodia (Chervier & Costedoat, 2017). Conversely, there are papers that either fail to discern a significant impact or even suggest detrimental outcomes, such as those observed in Bolivia (Wiik et al., 2019) and Sierra Leone (Wilebore et al., 2019). In addition, other studies have explored the impact of these projects on alternative outcomes, such as equity and well-being (Andersson et al., 2018; Naime et al., 2022), indicating that these effects are strongly influenced by information dissemination, as well as the enforcement of social and nonmonetary sanctions, both of which play a key role in shaping the mechanisms that drive the success or failure of these initiatives (Naime et al., 2022; Pfaff et al., 2019).

The evidence assessing the impact of collective PES on land-use changes is more limited, especially concerning projects on communal lands or within collective contractual frameworks. Research in Cambodia (Chervier & Costedoat, 2017; Clements et al., 2010), Ecuador (Cuenca et al., 2018) and Mexico (Alix-Garcia et al., 2012, 2018) suggests that initiatives deployed on communal lands are effective in curbing deforestation. Hayes et al. (2019) conducted a systematic review to understand how collective PES addresses the free-rider problem and second-order collective action challenges, analyzing studies that go beyond impact evaluation. The main conclusion of this review is that the effectiveness level is contingent upon the characteristics of the local institutional context.

Beyond the specific local conditions highlighted by Hayes et al. (2019), there is an increasing consensus that the success of collective PES is founded on three elements: community participation in the creation and clarification of norms for shared resource management; the effectiveness of monitoring and sanction systems; and the positive impact of these programs on poverty reduction and family income improvement (Brownson et al., 2019; Hayes et al., 2019). Additionally, Brownson et al. (2019) highlights that activities related to the negotiation and operational mode of collective PES provide a fertile ground to enhance social capital and boost community support over time. These components resonate with the common-pool resource literature as embodying principles of commons governance. As we are going to argue, this framework will be central in our argument that scenarios advancing collective REDD+ foster elements to reinforce some of the eight principles (Ostrom, 1990) for managing a commons.

Impact of REDD+ in Land Use. Unique within the PES framework, REDD+ is distinguished by specific attributes that could shape its impacts. One distinction is the funding source: Main Latin America REDD+ specific projects are predominantly financed through carbon market transactions, unlike traditional PES initiatives. This difference extends to REDD+'s operational mechanisms, characterized by stringent regulatory frameworks and rigorous monitoring protocols. Moreover, the financial compensation for environmental services under REDD+ is linked to carbon pricing, paying lower prices compared to PES projects. For instance, in Colombia, the compensation for preventing deforestation per hectare under a PES scheme varies between USD 140 and 860 annually. In contrast, with REDD+, using a conservative carbon credit price of USD 4, compensation could reach USD 48 per hectare per year (this is considering seven projects in Colombia).

Despite these institutional innovations, the evidence on REDD+ effectiveness remains mixed. Only two studies have analyzed the impacts of REDD+ across multiple countries, and their findings present contradictory results. Guizar-Coutiño et al. (2022) examined 40 REDD+ projects in nine tropical countries and reported a 47% reduction in deforestation within the projects' first five years of implementation. On the other hand, West et al. (2023) examined 26 projects in six tropical countries and stated that "the majority of the projects have not substantially decreased deforestation. For the few that have, the decline in deforestation is considerably less than what has been reported." Despite these differing findings, both articles agree that the emission reductions achieved are lower than what the projects claim for offsets in carbon markets. Studies questioning additionality and the risk of over-crediting highlight the need for a review of assessment methodologies, particularly the ex-ante baselines used by corporations for emissions claims (Badgley et al., 2022; Guizar-Coutiño et al., 2022; Haya et al., 2023; West et al., 2020).

At the subnational level, a limited number of studies find a causal relationship between REDD+ initiatives and deforestation. One notable study on a jurisdictional project in Guyana reported a 35% decrease in deforestation over five years, partially attributing this to the reduction in gold mining due to REDD+ payments, especially during a period of rising gold prices, which is a major deforestation driver in the area (Roopsind et al., 2019). In Brazil, four studies (Correa et al., 2020; Demarchi et al., 2023; Simonet et al., 2019; West et al., 2020) and one in Sierra Leone (Malan et al., 2024) provide varied findings that are limited in scope due to the small areas analyzed. Some showed significant reductions in deforestation (Demarchi et al., 2023; Simonet et al., 2019), while others (Correa et al.,

2020; West et al., 2020) reported minimal impact from the projects reviewed. Malan et al. (2024) also discovered a considerable effect on curbing deforestation but noted no change in economic wellbeing and conservation attitudes, making it the only study to assess these broader aspects. As Malan et al. (2024) observed, these investigations primarily focus on the effects of REDD+ on deforestation rather than its impacts on livelihoods or the behavioral mechanisms behind these effects.

Methodological choices explain part of the mixed results. Many of the influential studies cited above—including West et al. (2023), Guizar-Coutiño et al. (2022) and Badgley et al. (2022)—construct counterfactuals using matching or synthetic-control (SC) methods. Although these quasi-experimental approaches improve on simple historical baselines, they remain vulnerable to time-varying unobservables. Graham et al. (2023) make this critique in general terms: if a project area and its donor pool diverge in dynamic drivers—such as road building, agricultural prices or forest type—the SC weights no longer approximate the true counterfactual. We illustrate this problem with the Colombian projects analysed by West et al. (2023): their donor units lie in Vaupés (Amazonas), a region whose development trajectory differs substantially from the treated areas in Chocó (Pacific), potentially biasing the estimated impacts.

2.2 Illicit crops, drug policies and deforestation

Our work also contributes to the growing body of literature examining the impact of illicit crops as direct and indirect drivers of deforestation (Brombacher et al., 2021; Davalos et al., 2011; Dávalos, 2021; Erasso & Vélez, 2020; Marín-Llanes et al., 2024; P. J. Murillo-Sandoval et al., 2023; Quiroga et al., 2022) and the effects of state interventions and drug policy on the environment (Dávalos, 2021; Dávalos et al., 2016; P. Murillo-Sandoval et al., 2020; UNDP, 2023). This literature has not identified coca crops as a direct driver of deforestation in Colombia, but rather as an indirect one due to their spillover effects on agricultural sectors and productive transformations. Thus, this highlights the role of the coca economy as a catalyst for regional economic activity, which in turn could generate deforestation. Several studies have found that the so-called "war on drugs", particularly the war on cocaine, has not only failed to reduce supply and coca cultivation but has also led to unintended consequences. For example, aerial fumigation with glyphosate has proven to be both ineffective and cost-inefficient (Mejía et al., 2017), while also imposing disproportionate health (Camacho & Mejía, 2017) and environmental costs (Rubiano Lizarazo et al., 2020). Specifically, aerial spraying with glyphosate has contributed to deforestation by displacing coca crops (Rincón-Ruiz et al., 2013). Other strategies aimed at addressing the structural causes of poverty, such as alternative development projects that include conditional cash transfers, have also failed to reduce coca cultivation. Moreover, due to their agricultural focus and the displacement of coca crops, these initiatives have contributed to increased deforestation (DNP, 2023; Londoño et al., 2024).

In contrast, our contribution to this body of literature demonstrates that collective REDD+ agreements, designed solely as forest conservation tools and without conditioning the program on the absence of coca cultivation, have effectively slowed coca expansion. This finding substantiates the fact that well-crafted environmental incentives can influence land-use decisions, even within illicit economies. This evidence expands the discourse on drug

policy effectiveness, indicating that integrated conservation strategies can achieve social and ecological benefits that prohibitionist and alternative development programs have not secured. By documenting the first sustained indirect reduction of coca cultivation prompted by an environmental policy, we introduce a novel approach for synergistic governance in forest frontiers.

2.3 Collective Market-Based Approaches

In the framework of PES, one actor is willing to pay for the ecological benefits of conserved nature, and the other actor agrees to conserve in order to receive that payment. For the latter to choose conservation, the payment received must compensate for the opportunity costs they face when deciding to conserve. In REDD+, this means that revenues from the sale of carbon credits must be higher than or at least equal to the income received by forest inhabitants to use the land for agriculture, mining and, in particular, for logging (Engel, 2016; Engel et al., 2008), and, to grow coca crops[3]. Further, it is also important to bear in mind the variations in prices of commodities that are linked to intensive and extensive land use. As prices for agricultural products, minerals, and timber fluctuate—commodities that often drive deforestation—the opportunity costs for forest inhabitants are not constant. This price variability is a factor that adds complexity to PES and REDD+ contracts (Angelsen, 2014). However, even with this challenge, this mechanism is clear for a PES or REDD+ contract signed by the direct provider of the service; in this case, the farmer receives direct compensation.

But what happens in cases where forests and ecosystems are collectively owned? How does the logic of compensation work when a collective sells carbon credits? In these cases, the income farmers previously received from illegal logging or coca leaves has no direct compensation or immediate relationship with the payments for carbon credits, as they are collective. Given that the decision on how to distribute collective payments is also collective, this can imply the absence of individual payments and, consequently, the absence of compensation to *corteros* (loggers) and for coca growers. Even in hybrid compensation schemes, where part of the collective payments goes to a common fund and the rest is distributed between each individual of the community, the compensation does not necessarily cover the cost of protecting the forest by stopping agricultural activity.

In this context, collective REDD+ projects, like any collective PES, face two challenges. The first is when collective payments fail to offset the income forgone by abandoning unsustainable practices. The second challenge is the first-order collective action problem, where the individual compensation from collective PES is only partly conditioned on individual behaviour (Hayes et al., 2019). A single farmer has the option to not cooperate in the collective effort (by persisting in growing coca) and still receive the advantages of the collective payments and investments. Additionally, "achieving collective REDD+ outcomes is difficult, particularly when the user group is large, heterogeneous and poor, and the forest benefit flow and economic environment unstable" (Angelsen, 2009).

Given these two challenges coupled with the lack of a direct relationship between changes

[3]It is important to clarify that, according to Colombian regulation on PES, opportunity costs for illegal activities, such as coca cultivation, cannot be covered (Decree 870 of 2017).

in farming practices, compensation, and the persistent pressure of coca cultivation in collective territories, one question emerges: can REDD+ projects help to control coca expansion and deforestation?. Below we discuss four plausible mechanisms: governance and social capital, individual and collective benefits, the role of intermediaries, and surveillance and global climate governance, that could address these challenges, considering the interplay between collective governance, economic incentives, and the behavioral dynamics within these communities.

3 Context

In Colombia, a country where approximately 50% of its territory is covered by forests, the deployment of REDD+ initiatives has emerged as a national priority in the fight against climate change. Two factors have played an important role in this prioritization. First, since 2017, regulations have been established that allow the national carbon tax to be offset with carbon credits from the voluntary market, creating significant domestic demand (Schmid & Castro, 2023). Second, the Rights and Resources Initiative has highlighted Colombia as one of only three countries (along with Brazil and Costa Rica) where carbon rights are clearly linked to land or forest ownership. This linkage is crucial as it empowers communities holding land rights to potentially develop and benefit from such projects autonomously. Due to this favorable environment, Colombia's international participation in REDD+ projects is noteworthy. In terms of the number of projects, Colombia ranks second globally only to Brazil, followed by Indonesia (Haya et al., 2023). With regard to the volume of credits, Peru, Indonesia, Brazil, and Colombia are at the forefront. From 2011 to 2021, Colombia sold more than 26 million carbon credits, generating a total of roughly 104 million dollars (Haya et al., 2023).

A defining characteristic of REDD+ implementation in Colombia is its concentration within collective territories. Since 2009, REDD+ projects have been largely implemented in lands belonging to both Indigenous peoples and Afro-Colombian communities. By 2022, over 100 REDD+ projects had been registered with the National Registry for the Reduction of Greenhouse Gas Emissions (RENARE)[4]. Out of these, 64 projects are classified as collective REDD+ initiatives, involving Indigenous communities (46 projects) and Afro-descendant communities (18 projects).

This study focuses specifically on the latter group within the Pacific region. Here, Afro-Colombian communities hold collective ownership over approximately 5.7 million hectares. The legal framework for this collective ownership was established by Law 70 of 1993, which also introduced Afro-Colombian Community Councils (CCs) as the social organizations tasked with governance over these lands[5] (Peña et al., 2017; Vélez et al., 2020). Currently, 178 CCs have been formally recognized with land titles in the region. Leveraging the carbon

[4]This registry, established by Resolution 1447 of 2018, was designed for the management of Greenhouse Gas GHG mitigation initiatives at the national level and remained open for public consultation until August 2022

[5]CCs are local governance entities established under Law 70 of 1993, which granted Afro-Colombian communities collective ownership rights over land. These councils are tasked with managing collective territories, including decisions on the use of natural resources and the regulation of external actors.

rights associated with their land ownership, many CCs have engaged with the voluntary carbon market through REDD+ projects.

Although these collective territories, including those managed by CCs, have historically shown varying levels of success in mitigating deforestation (Bonilla-Mejía & Higuera-Mendieta, 2019; Romero & Saavedra, 2020; Vélez et al., 2020) and controlling illicit coca cultivation (Lobo & Vélez, 2022), conservation remains a costly endeavor facing significant pressures. Since approximately 2014, rates of deforestation within these collective territories have alarmingly increased, in some cases exceeding those observed in National Natural Parks and Indigenous reserves (Castro & WWF, 2023). Simultaneously, coca cultivation has expanded, with particularly high concentrations in the Pacific region (United Nations Office on Drugs and Crime, 2023). These trends highlight how inherent governance challenges in collective tenure—such as high monitoring costs, potential free-riding problems, and managing the opportunity costs associated with conservation—can limit the effectiveness of community-based conservation efforts (Hayes et al., 2019; Naime et al., 2022).

Knowledge about the actual performance and impacts of collective REDD+ projects in Colombia remains limited, with few academic analyses available, notably some focusing on the Colombian Amazon (Schmid & Castro, 2023; SINCHI, 2023). Furthermore, journalistic investigations have raised serious concerns regarding the integrity and implementation of several REDD+ projects across Colombia. Issues identified primarily in the Amazon region and Nariño include allegations of inflated deforestation baselines, incomplete community consultations, misrepresentation, unauthorized agreements, legal challenges, double accounting, community exclusion, problematic developers, and lack of community involvement (Bermúdez, 2021, 2022a, 2022b, 2023a, 2023b). These documented problems span governance, legal, and ethical domains, casting doubt on the effectiveness and fairness of some REDD+ initiatives in the country (Schmid & Castro, 2025).

In fact, the Colombian Constitutional Court, in its Sentence T-248 of 2024, responded to one such case by recognizing systematic failures in socialization and free, prior, and informed consent processes. The Court assigned specific responsibilities to public institutions to ensure adequate consultation, create clear policy guidelines for REDD+, and guarantee effective community participation. However, it is crucial to note that these highly publicized cases and the court ruling primarily concern regions distinct from the Colombian Pacific, potentially involving different intermediaries and institutional settings. Acknowledging the complexities surrounding REDD+ implementation, we must also recognize the limitations of our study due to its lack of in-depth qualitative analysis, which would be important to understand why the majority of projects in the Pacific region don‘t face similar issues and how CC governance structures have mitigated these risks.

Our study is centered on the Pacific region, including only collective titles of Afro-Colombian communities. Between 2009 and 2020, 33 CCs signed 17 REDD+ contracts (Figure 1 and Figure A.1). The first REDD+ project from collective councils dates back to 2009. By the end of 2013, eight REDD+ projects from CCs had emerged under the BIOREDD+ program. [6] The projects involved in this program signed eight agreements with Fondo Acción to act as their representative and intermediary in the negotiation of

[6]This was a cooperation program from USAID initiated in 2012 to bolster community capacities for engaging with REDD+ investments.

carbon credits. After that, another 10 projects were signed with other intermediaries more integrated with the carbon market.

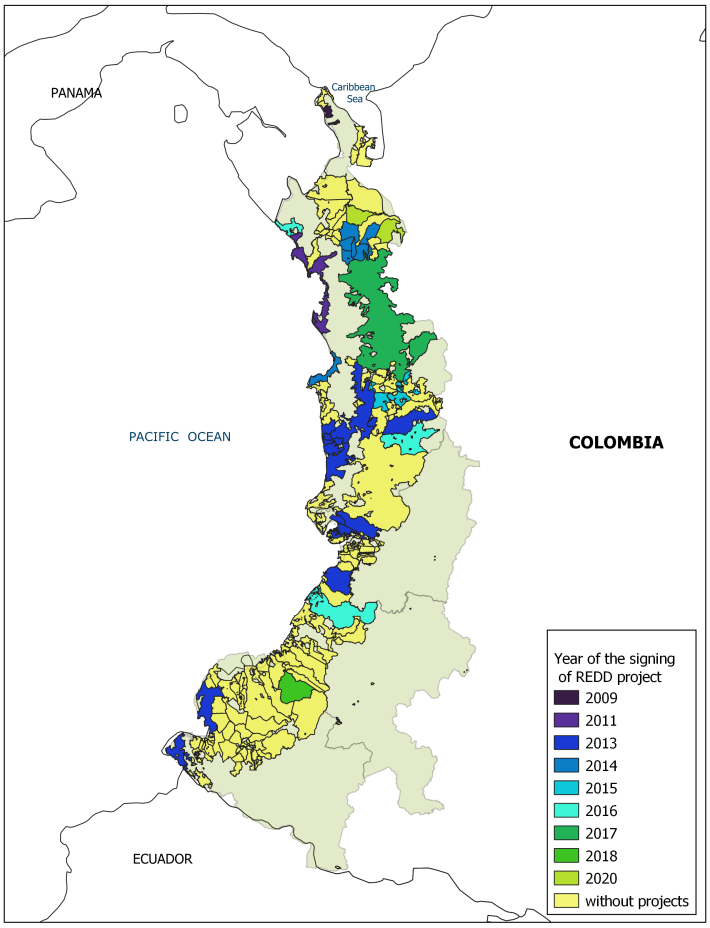


Figure 1: Geographical distribution of treated and control community councils.

Notes: The map illustrates the geographical distribution of CCs in the study region. Yellow represent non-treated zones. The green background represents the broader geographical context of the region within the Pacific coast.

Figure 1 illustrates the geographical distribution of communities that sign a REDD+ project, categorized by the year of the treatment, along Colombia's Pacific coast. The map reveals that treated regions are scattered throughout the area.

Most agreements in this region prioritized collective payments over individual compensation, channeling resources toward investments in public goods and community development. As Hayes et al. (2019) suggests, this preference probably reflects higher levels of trust in the region and the strong collective governance. In contrast, in contexts where trust in community leadership is low, households often favor individual payments for their transparency and ability to meet personal needs. This characteristic of collective contracts in the Pacific region could represent a key distinction from those in Indigenous collective communities in the Amazon, where governance structures and trust dynamics may differ, and some of them

decided to give individual payments.

Figure A.2 illustrates key phases in the life cycle of a REDD+ project. The signing of the agreement marks the beginning of the project's operational phase, which is also considered the initiation of the treatment. From this point on, coordinated efforts are launched, including the development of the project's planning document, which is essential to achieve deforestation control targets set for future verification.

Approximately two to three years after the agreement is signed, the first verification process begins [7]. Three years later, the first harvest takes place, allowing carbon credits to be issued and sold on the market. Once the credits are sold, the revenue starts flowing to the community. In the timeline example presented in A.2, it is estimated that by 2018 - five years after the agreement was signed - the funds were able to reach the communities. In subsequent years, the cycles of verification, harvest, and credit sales repeat, ensuring continued financial inflows.

Each REDD+ project must formulate a development and implementation plan with the activities to be carried out to achieve the emission reduction goals and the amount of carbon credits to be issued for validation by a certifier. After reviewing in detail 8 of the BIOREDD projects to which we have access, the following elements have been identified as common characteristics. Central to the initiative is the strengthening of land tenure and forest governance, which is achieved through a collaborative process that includes the dissemination of by-laws during community meetings, and the development and implementation of sustainable forest and land use management plans. These plans are updated to align with community priorities and are formally approved by community assemblies, ensuring local support and facilitating enforcement. A key element is the delimitation of significant reserve areas as conservation zones, often accompanied by the participation of local *corteros* and *piangueras* (women mollusk gatherers in mangrove zones) in reforestation and regeneration activities within these reserves. Community involvement is further enhanced by actively patrolled zones, ensuring the adherence to land use rules and increasing local awareness of land use zones.

Furthermore, the qualitative assessment of the projects in the Pacific region highlights key areas of impact that align with the mechanisms proposed in our analysis (Acción, 2020; USAID, 2022): improvements in local governance through strengthened community-based decision-making structures, enhanced land tenure security, and more effective benefit-sharing arrangements. In terms of livelihoods, communities have diversified their productive activities by adopting alternative income-generating strategies such as cacao and banana cultivation, silvopastoral systems, and community-managed fishing initiatives. Leadership development is also an outcome, with targeted efforts to empower women and youth, foster inclusive participation, and build local leadership capacities. Additionally, the projects have contributed to social investments, including the provision of educational scholarships, improved health infrastructure, and the promotion of cultural and recreational activities. These findings underscore how REDD+ interventions have not only reinforced local governance and diversified livelihoods but also catalyzed broader social benefits.

Lower estimates suggest that communities receive $48 USD per hectare per year from

[7] verification is an independent audit by an accredited validation/verification body (VVB) that scrutinises monitoring data and, if satisfactory, clears the registry to issue carbon credits

REDD+ projects, which is considered low compensation. [8] However, it is important to mention that, different from Indigenous resguardos, Afro-Colombian communities do not receive financial transfers from the government; thus, this may represent the first time they have access to resources and autonomy over them to finance public goods for the community. Over the past five years, 14 projects have collectively received an estimated total of 3 million dollars (Haya et al., 2023).

4 Emprirical Strategy and Data

This section explains the main data sources used for the analysis and describes the trends of the primary outcome variables. It also outlines the empirical strategy and the identification assumptions.

4.1 Data and Descriptive Statistics

This subsection outlines the data sources used in the analysis and presents descriptive statistics to contextualize key variables.

REDD+ Data. According to RENARE, from 2009 to 2020, 33 CCs(40% of CC land in pacific region) signed 17 REDD + contracts (see Appendix: Table A.1) with the expectation of generating over 42 million carbon credits over 30 years. We only have access to the names of community areas, which we matched with the polygons for each CC. It should be noted that we do not have access to the specific polygons where REDD+ implementation occurs. Each CC designates a subarea within their territory responsible for emission reductions; not all the territory is involved, as the communities reserve some areas for agriculture, legal logging, and subsistence activities.

Coca Data. We use the coca cultivation census published annually by the United Nations Office on Drugs and Crime (UNODC) for Colombia since 2001. This census contains geo-referenced information on the estimated area of coca cultivation in each 1 km^2 polygon. Information is available for the entire country over a 20-year period. We select all polygons inside the Pacific region and identify to which CCs they pertain.

Deforestation Data. Our research uses deforestation data derived from raster datasets previously processed by the Institute of Hydrology, Meteorology, and Environmental Studies (IDEAM). The data are categorized into five distinct classes with a spatial resolution of 30m x 30m per pixel: 1 = stable forest, 2 = deforestation, 3 = unavailable data, 4 = regeneration, and 5 = unstable forest. For computational efficiency, we only processed category 2 (deforestation corresponds to the direct conversion of natural forest cover to another type of cover in a given period of time)) and upscaled the data from 30x30m pixels to 90x90m. Upscaling was performed by aggregating neighboring pixels into larger grid cells. That is, values from adjacent 30×30 m pixels were summed within a 90×90 m cell, effectively combining information from multiple smaller units into a single coarser one. This upscaling process allows us to manage the data more effectively. The IDEAM raster data span 15 years, during which we pinpoint deforestation levels within 90x90m grids across areas managed by

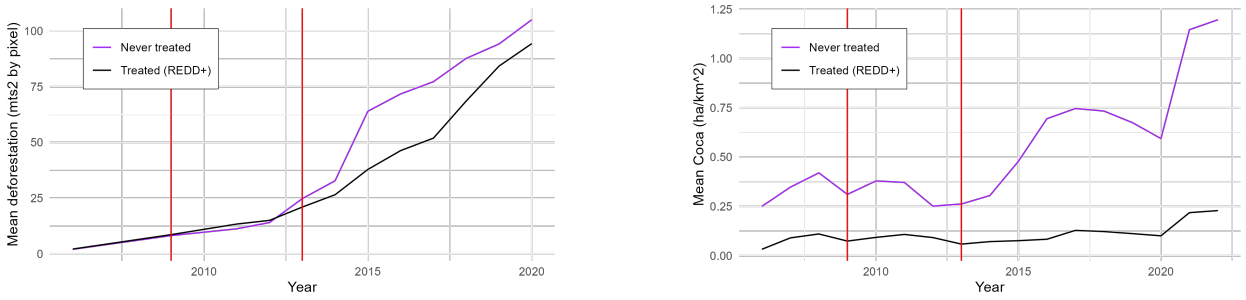
[8] Some estimates suggest that an area of coca generates 14 million Colombian pesos annually, equivalent to approximately 3,712 USD per hectare per year.

CCs in the Pacific region. We end up with a panel dataset comprising over 7 million pixels, each with 15 years of data. Given the size of the panel (more than 100 million observations), to run the estimations, we extracted a stratified sample of 5% (we test the robustness with other random samples). Stratification was performed at the community council level. For robustness, we also processed the Global Forest Change dataset (Hansen et al., 2013), but our primary dataset is from IDEAM, as it is more reliable in controlling for cloud cover and accounts for the specific conditions of the Pacific region.

Figure 2 presents the annual trajectory of the outcome variables for CC with and without REDD+ project. After 2012, a clear divergence in trends in deforestation and coca cultivation emerges, becoming more pronounced after most CC adopted REDD+ projects. The divergence is particularly evident in coca cultivation from 2014 onward, while deforestation trends start to shift as early as 2012.

Since our empirical model relies on comparing treated and not-yet-treated areas, we present descriptive statistics for deforestation and coca crops by treatment cohort in Table A.2 and Table A.3 in the appendix. For accumulated deforestation (Table A.2), areas never engaged in REDD+ projects exhibit one of the highest mean deforested areas. In particular, areas in the 2015 and 2020 cohort show the largest mean accumulated deforestation among treatment cohorts, indicating potentially higher baseline pressures for these cohorts.

For coca cultivation (Table A.3), areas never treated also show the highest mean coca extent (0.54 ha/km^2). In contrast, treatment cohorts exhibit minimal coca presence, except for the 2013 and 2016 cohort. In general, we can see that areas engaging in REDD+ projects often faced distinct pre-treatment levels compared to those never involved.



(a) Deforestation

(b) Coca

Figure 2: Trajectories of deforestation and Coca Crops in CCs with and without REDD+.

Notes: The figure illustrates the trajectories of deforestation/coca cultivation for CC with and without REDD+ projects. The purple line represents CCs treated with REDD+, while the black line corresponds to untreated CCs. The vertical red lines mark key intervention years. Data were aggregated at the community level, and trends were based on weighted annual averages.

4.2 Empirical Strategy

To estimate the effect of the signing of REDD+ contracts on coca crops and deforestation, we implement a difference-in-differences (DiD) strategy with multiple time periods, following the approach developed by Callaway and Sant'Anna (2021). This allows us to compare outcomes before and after the signing of REDD+ contracts within community councils (CCs),

accounting for the staggered timing of contract adoption. The observation unit is the polygon p within a community council c, and our outcome variable $Y_{p,c,t}$ represents either the accumulate deforested area (m^2) or the area of coca cultivation (km^2) in polygon p during period t. Formally, we estimate the following equation:

$$Y_{p,c,t} \;=\; \alpha_c \;+\; \alpha_t \;+\; \sum_{e} \sum_{l \neq -1} \tau_{e,l} \Big[\mathbf{1}\{E_c = e\} \;\times\; \mathbf{1}\{t - e = l\} \Big] \;+\; \epsilon_{p,c,t}, \tag{1}$$

where $Y_{p,c,t}$ denotes the outcome of interest for polygon p within council c at time t. The parameters α_c and α_t are council and time fixed effects, respectively. α_c absorbs all time-invariant heterogeneity across community councils, while α_t captures any common shocks or time trends affecting all polygons in a given period t. The variable $E_c = e$ indicates that council c is treated for the first time in period e, and the event time $l = t - e$ measures the distance from the onset of treatment. $\tau_{e,l}$ captures the causal effect in event-time l for councils whose treatment began in period e, relative to that baseline. Finally, $\varepsilon_{c,t}$ is the idiosyncratic error term. After estimation, we aggregate event-time coefficients $\tau_{e,l}$ to construct average treatment effects across cohorts and post-treatment years. We also generate an event-study figure showing the dynamic impact of REDD+ contracts over time relative to the signing event.

For causal identification, our analysis relies on two key assumptions. The first is parallel trends, which assumes that, in the absence of REDD+ contract signing, trends in coca cultivation and deforestation in treated councils would have evolved similarly to those in not yet treated councils. While it is impossible to directly test this assumption, the absence of pre-trends in our staggered difference-in-differences model provides supporting evidence for its plausibility (see: Figure 3).

The second is no anticipatory behavior, which implies that CCs did not adjust their behavior (e.g., in deforestation or coca cultivation) in anticipation of signing REDD+ contracts. These assumptions are critical to ensuring that our difference-in-differences estimates capture the causal effect of REDD+ contracts rather than preexisting differences or strategic adjustments by councils.

To address the heterogeneity inherent in treatment timing, we employ the staggered difference-in-differences framework of Callaway and Sant'Anna (2021). This approach estimates cohort-specific average treatment effects by comparing outcomes for councils treated in a specific period e (the cohort) with outcomes for councils that are not yet treated during the same time period t. By explicitly accounting for variation in treatment timing, this method isolates the causal impact of REDD+ contracts more reliably. Furthermore, it directly addresses the potential problem of negative weighting that can bias standard two-way fixed effects estimators when treatment effects are heterogeneous across cohorts or over time (see: Callaway and Sant'Anna (2021), de Chaisemartin and D'Haultfœuille (2020), and Roth et al. (2023)). The CS method flexibly accommodates heterogeneous treatment effects both across different treatment cohorts (e) and across time relative to the treatment event (l).

Our primary control group consists of not yet treated units—CCs that have not signed REDD+ contracts during the observation period. This group serves as a natural comparison for councils that have already signed contracts, as they share similar institutional and environmental contexts but have not yet been exposed to the intervention. Additionally, we

perform robustness checks using never treated units as an alternative control group. These units, while not participating in REDD+, share several key characteristics with treated councils, such as similar socioeconomic conditions, exposure to illegal groups (Peña et al., 2017), and a history of reduction in deforestation relative to non-titled areas in the region (Romero & Saavedra, 2020; Vélez et al., 2020).

A potential concern in our analysis is the possibility that the timing of REDD+ contract signing may introduce bias if CCs that sign earlier differ systematically from those signing later or not participating at all. For instance, councils with stronger governance structures or higher initial deforestation rates may have been more likely to sign contracts earlier and follow a different path than those that signed later. To address this, the staggered difference-in-differences strategy inherently controls for these differences by estimating treatment effects specific to each cohort and period.

5 Results

This section presents our results on the impact of REDD+ on deforestation and coca cultivation. Each subsection first outlines the main findings, followed by robustness checks, and finally, we analyze potential heterogeneities to highlight factors that may explain impact variations.

5.1 Impact on Deforestation

Table 1 presents the average treatment effects of the dynamic event study using three estimators: Callaway and Sant'Anna (2021), Wooldridge (2021), and Sun and Abraham (2021). We find an average reduction of 33 m2 per pixel in deforestation following the signing of a REDD+ agreement. This corresponds to an 33.34% decrease relative to the average deforestation observed in all the pixels in CCs for the entire study period (103 m^2 per pixel). For the 2,339,366 ha (2,888,107 grid cells of 8,100 m^2) in CCs that signed REDD+ projects, this reduction translates into approximately 9,695 hectares per year of avoided deforestation. If all 5,716,302 ha (7,057,163 grid cells) in CCs across the Pacific adopted REDD+ projects, the potential avoided deforestation could reach 23,691 hectares per year.

The temporal dynamics, illustrated in Figure 3, shows that the estimates for years prior to the signing are equal to zero, suggesting that we are correctly identifying a causal effect. Starting from the first year post-treatment onward, we observe a reduction in deforestation with significant estimators. These findings emphasize the role of REDD+ projects in effectively containing and reducing deforestation pressures, particularly in regions with pre-existing deforestation challenges (see event study plot for other estimators in Appendix Figures A.6a, A.6b). .

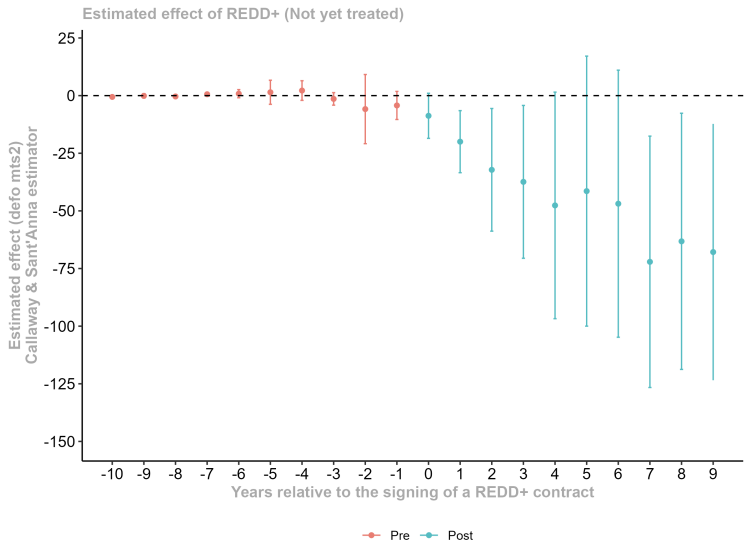


Figure 3: Event Study: Impact of REDD+ on Deforestation (C&S 2021)

Notes: The figure shows coefficients and 95% confidence intervals from event study regressions using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No additional covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is a pixel-year.

Table 1: Impact on Deforestation

	C&S	Wooldridge	S&A
ATT	-33.57***	-39.39**	-26.90**
Std. Error	(12.97)	(18.48)	(12.97)
N	5,294,250	5,294,250	5,294,250
Mean 2006-2021	103.04		

Notes: This table presents the results of difference-in-difference staggered regressions estimating the average impact of REDD+ signature on deforestation. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No additional covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year. The estimation was done with the R `did`, `etwfe`, and `sunab` packages.

The avoided deforestation can also be translated into emissions reductions. To put this into perspective, we use two years as examples to calculate the magnitude of the impact. Colombia registered 219,552 hectares of deforestation in 2017 and 174,103 hectares in 2021. The deforestation avoided by REDD+ projects in treated areas represents between 4.42% (2017) and 5.57% (2021) of the total deforestation recorded in those years.

In terms of emissions, given that deforestation in the Pacific region releases between 95 and 114 tons of CO_2 per hectare, the avoided deforestation corresponds to emissions reductions of approximately 921,025 to 1,104,330 tons of CO_2 in treated areas for 2021. If all CCs across the Pacific implemented REDD+ projects, the potential avoided deforestation could account for between 13.61% (2021) and 10.79% (2017) of the national total deforestation, with emissions reductions ranging from 2,250,645 to 2,700,774 tons of CO_2. The range of 2.25 to 2.7 million tons of CO avoided represents between 1.1% and 1.3% of Colombia's total annual emissions. However, considering only emissions from land-use change, this reduction would represent between 3.46% and 4.15% of the national total from deforestation.

5.1.1 Robustness Checks

The observed decrease in deforestation following the signing of a REDD+ project remains robust across a series of robustness tests. Specifically, the results are valid under the use of different samples, an alternative data source, an alternative control group and as we already presented previously to different estimators. We also check the results, excluding the biggest councils.

Due to computational constraints, the results presented in Figure 3 are based on a randomized sample stratified by community council. To verify that sub-sampling is not driving our results, we replicate the analysis on three independent stratified draws: (i) two additional 5% samples and (ii) one 10% sample. Figure A.3a, A.3b, A.3c plots the corresponding event-study coefficients; point estimates and their 95% confidence bands are virtually indistinguishable across draws. These checks confirm that our conclusions are not an artefact of the sampling scheme.

Appendix Figure A.4 presents an additional check using an alternative dataset, the Global Forest Change (GFC) database (Hansen et al., 2013). The results of the dynamic event study reveal the same trend, with the key difference being that all posttreatment coefficients are significant and negative, with slightly higher magnitudes. Because GFC is produced independently of IDEAM's national monitoring system, this convergence rules out the possibility that our findings are an artefact of country-specific processing choices or measurement error in the IDEAM raster. In other words, the effect survives a cross-validation against a global, externally generated benchmark.

The main specification employs not-yet-treated pixels as a control group. A less restrictive approach includes other CCs that never (until 2022) signed REDD+ projects. When incorporating never-treated pixels, the results remain consistent (see Appendix Figure A.5a). Finally, we pool the two sets of controls—never-treated and not-yet-treated pixels—into a single, larger comparison group. This almost triples the number of untreated observations and again delivers coefficients that overlap the baseline at every relative period (Appendix Figure A.5b). Hence, our results are robust to any of the three plausible control-group definitions.

Finally, to ensure that the observed effect is not driven by a specific council, another robustness check replicated the analysis while excluding one community council at a time, and the results remained stable (Appendix Figure A.13). For this test, we use eight CC with the biggest extension areas.

In sum, every robustness probe, alternative estimators, four independent stratified sub-

samples, an external forest-loss dataset, three control-group definitions, and leave-one-council-out tests, points in the same direction and confirm our main results. Collectively, these exercises rule out the main threats to inference, sampling variability, measurement error, model dependence, and influential observations, and give us strong confidence that the documented effect is genuine rather than an artefact of data choice, estimator, or sample composition.

5.2 Impact on Coca Crops

Table 2 presents the main results on coca for the three estimators Wooldridge (2021) and Sun and Abraham (2021) estimators. In all cases, we find negative and significant effects of roughly the same magnitude. On average, we estimate a reduction of 0.33 ha/km^2 in the area cultivated with coca following the signing of a REDD+ contract. As the event study in Figure 4 shows, since the signing period, we observe negative and significant effects on coca cultivation (See Appendix Figure A.12 Event study plots for the others estimators). The magnitude of this effect is notable, representing a 90.46% reduction relative to the average coca cultivation for all the grids between 2006 and 2021 (0.367 ha/km^2). In terms of total impact, for the 21,509 km^2 corresponding to the grid cells in CCs that signed REDD+ projects, the estimated reduction or containment in coca cultivation is approximately 7,098 hectares per year. Furthermore, if we extrapolate this measure to all 54,348 km^2 of grid cells in CCs across the Pacific, including the never-treated areas, the potential reduction in coca cultivation could reach approximately 18,013 hectares per year. To put the figure in perspective, 18,000 ha correspond to roughly 7% of Colombia's total coca area in 2023, almost 90% of the government's annual manual-eradication target for 2023, and about one-third of all coca currently found in Afro-Colombian collective lands.

The results indicate that REDD+ projects significantly contain the growth rate of coca crop extension in treated areas. While non-treated areas experienced substantial increases in coca cultivation, the presence of REDD+ projects acted as a containment mechanism, mitigating the expected expansion. This underscores the effectiveness of REDD+ projects in attenuating the upward trend observed in non-treated areas.

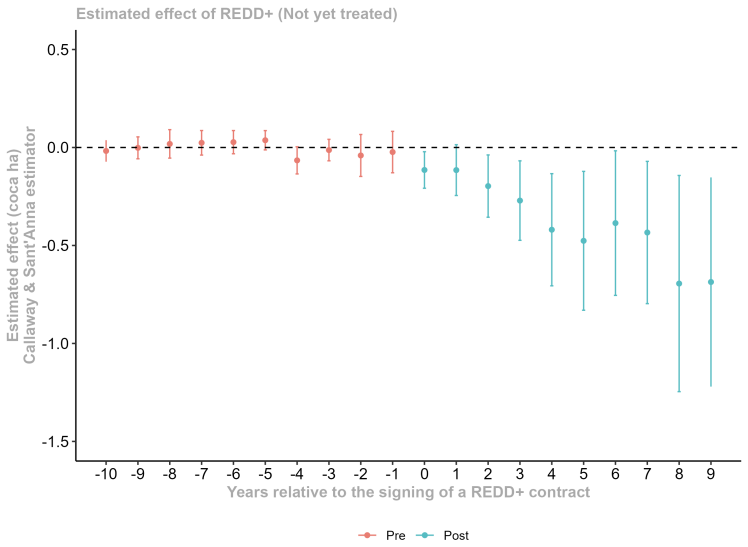


Figure 4: Event Study: Impact of REDD+ on Coca Crops (C&S 2021)

Notes: The figure shows coefficients and 95% confidence intervals from event study estimates of the effect of REDD on coca crops using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is coca in ha at 1km^2 pixel resolution. No additional covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Table 2: Main effects: Coca crops

	C & S	Wooldridge	S & A
CATT	-0.332***	-0.342***	-0.292***
Std. Error	(0.102)	(0.13)	(0.079)
N	1,195,656	1,195,656	1,195,656
Mean 2006-2021	0.367		

Notes: This table presents the results of difference-in-difference model estimating the average impact of REDD+ signature on coca crops. The outcome variable is coca in ha per km^2. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is a pixel-year. The estimation was done using the R `did`, `etwfe`, and `sunab` packages.

5.2.1 Robustness Checks: Coca Crops

The observed reduction in coca cultivation following the signing of REDD+ projects is supported by a series of robustness checks. The primary specification uses not-yet-treated pixels

as the control group. As a robustness check, we tested the results by including the never-treated group and a combination of both not-yet-treated and never-treated pixels. The results remain consistent, showing a reduction in coca cultivation post-treatment (see Appendix Figure A.11a). As mentioned previously we also applied two alternative approaches: Wooldridge (2021) and Sun and Abraham (2021). Both methods confirm the robustness of the findings, showing consistently negative and statistically significant post-treatment effects (Appendix Figure A.12a, A.12b).

Finally, to ensure that the results are not driven by any single Community Council (CC), the analysis was repeated while systematically excluding one treated CC at a time and the results remained stable (Appendix Figure A.13a). For this test, we chose eight CCs with the largest extension areas.

Once again, the containment effect proves robust after addressing the key threats to inference: control-group choice, estimator selection and influential observations.

5.3 Heterogeneous Effects

Potential sources of variation in the effect of REDD+ projects on deforestation and coca crops in CCs are now considered. The focus is on heterogeneous effects related to deforestation/coca prior to the signing of REDD+, the size of the area of the CC, and the intermediary with whom the project is signed.

We estimated the effect in different subsamples of interest to reveal heterogeneities in the effect of the treatment. We calculated the effects considering the role of different intermediaries in the signing of REDD+ contracts. Figure A.8 for deforestation and Figure A.14 for coca crops in the Appendix contain the results for three of the most important intermediaries in the carbon market in the Colombian Pacific. In this case, the control group consists of the never-treated units.

The analysis reveals small variations in the reduction of deforestation depending on the intermediary. For all intermediaries, the effects are consistently negative. For Fondo Acción, the effects are consistently negative and statistically significant, with a clear downward trend in deforestation rates three to five years after the start of the REDD+ contract. In contrast, for Biofix and Terra, while the coefficients remain negative, statistical significance is less consistent, likely due to reduced estimation power from sample partitioning. Similarly, The analysis of heterogeneous effects on coca crops shows limited variation across intermediaries. For Fondo Acción, the effects are consistently negative and statistically significant, indicating a steady reduction in coca density starting three to five years after the signing of the REDD+ contract. Biofix follows a similar trend, with negative effects becoming more pronounced over time, albeit with slightly wider confidence intervals. In contrast, for Terra, while the post-treatment effects are negative, statistical significance is weaker.

The heterogeneous analysis suggests that projects signed with Fondo Acción are more successful in achieving positive results from the early stages of implementation. This could be attributed to their ability to mobilize more resources during the years prior to the point at which the CC can begin selling carbon credits, providing stronger financial and operational leverage during critical early phases. As revealed by a community member involved in a project with this intermediary, one key element that helped transition away from unsustainable practices before receiving carbon credit revenue was Fondo Acción's integration

with other cooperative projects. These additional initiatives not only provided alternative income streams but also supported broader community processes, strengthening collective governance and capacity building.

To examine whether the observed effect is influenced by councils with stronger historical management of common resources and control of illegal activities, we conducted a heterogeneous analysis by splitting the sample based on median values of deforestation levels, presence of coca crops before REDD+ implementation, and community council area.

The heterogeneous analysis by pre-treatment deforestation levels reveals small differences. For both low and high deforestation pre-treatment areas (Appendix Figure A.9), the effects are consistently negative and statistically significant. The main distinction lies in the magnitude of the effects: for high deforestation pre-treatment areas (Panel b of Appendix Figure A.9b), the impacts are significantly larger, suggesting a stronger effect of REDD+ in containing pressures for deforestation these contexts.

This result is particularly important as it addresses one of the main criticisms of carbon offset contracts, the issue of additionality. A frequent critique is that these projects are not sufficiently concentrated in areas at high risk of deforestation. However, the evidence here confirms that in the case of these projects, REDD+ interventions are indeed effective in zones with greater deforestation risks, demonstrating their capacity to deliver additional benefits.

The heterogeneous analysis for coca reveals that in low coca pre-treatment areas (Panel A of Appendix Figure A.15a), there is an increase in coca density. In contrast, in high coca pre-treatment areas (Figure A.15b), the results show a reduction in coca density. This suggests that REDD+ projects were more effective in controlling both deforestation and the expansion of coca crops in territories facing greater pre-existing threats.

The final heterogeneous analysis examines the size of the CC and its influence on the effectiveness of REDD+ projects. Based on literature on commons, larger areas under collective governance often face challenges in monitoring and enforcing agreements, potentially weakening outcomes. The results reveal some differences in the impact of these projects depending on the territorial scale of the CC. Contrary to expectations, the effects on deforestation are higher for larger CCs, showing consistently negative and statistically significant impacts on deforestation (Panell B Appendix Figure A.10b). This highlights a unique feature of REDD+ projects—their ability to adapt to the inherent challenges of large collective territories. These projects appear to incorporate mechanisms that mitigate the risks of weak monitoring and enforcement across extensive areas, suggesting an effective design tailored to the complexities of larger councils.

The analysis of coca crop dynamics reveals that in small communities, the magnitude of the effect on coca reduction is higher than in large CCs (Figure A.16a, A.16b). However, coca density remains consistently negative and statistically significant across both cases, albeit slightly smaller in magnitude for larger councils. This finding highlights that REDD+ projects effectively curb coca expansion, even in larger community territories.

6 Discussion

Considering the collective action dilemmas and challenges previously presented, the following outlines a series of plausible mechanisms that explain the positive outcomes in containing the increase in coca cultivation and deforestation. In this section, we will discuss the potential mechanisms underlying these outcomes, categorized as follows: (1) governance and social capital, (2) individual and collective benefits, (4) the role of intermediaries, and (5) surveillance and global climate governance.

Governance and social capital. A fundamental mechanism driving the success of REDD+ projects is their capacity to reinforce local governance structures and collective action. Lobo & Vélez (2022) identified that the sustainability of resistance to illicit activities in community territories depends on four key factors: (a) the stock of social capital that determines organizational capabilities, (b) the specific normative content with which this capital is infused, (c) the legitimacy and influence of leadership, and (d) synergies among different levels of grassroots community organization. REDD+ programs directly engage with these elements, strengthening pre-existing governance systems while fostering new dynamics of collective management and a renewed web of nested organizations around different productive activities for the long-term well-being of the community.

REDD+ projects require high levels of coordination for negotiation and deliberation around the decision to sign and the types of rules to follow when a community signs a contract, this exercise plays a central role in this governance reinforcement. These processes activate and strengthen social capital, creating a renewed sense of collective responsibility and organization, and as Hayes (2019) notes, this depends on the quality of the previous system of governance. Moreover, the contract signing process involved developing plans that revisited norms concerning external actors, collective territorial development strategies, and frameworks for monitoring and accountability. In the specific case of coca cultivation, communities that have previously managed to limit coca expansion in their territory have been able to activate and adapt these mechanisms effectively within the REDD+ framework. This dynamic addresses both the primary collective action dilemma—the risk of free-riding—and the secondary dilemma, which involves achieving agreements and implementing effective governance (Hayes et al., 2019).

REDD+ projects also enable communities to assert greater control over their territories. This is particularly relevant in regions where external actors, such as illegal loggers and coca growers, threaten local stability. However, it is important to note that not all loggers or coca growers are external actors; in some cases, they are community members. In these instances, REDD+ projects reinforce collective agreements and social norms, ensuring that neighbors hold each other accountable when everyone has committed to a contract. Furthermore, when coca cultivation is driven by external actors that might exploit governance gaps, REDD+ projects revitalize social capital processes and redefine territorial boundaries. This revitalization empowers councils to exercise stronger control over who enters their territories, effectively reducing illicit activities like coca cultivation and illegal logging. The reactivation of community-based monitoring mechanisms ensures that governance improvements apply not only to coca but also to other external pressures that threaten ecosystems and local livelihoods.

Finally, these projects have facilitated the creation of a variety of suborganizations within the CC, aligning with one of the eight principles of Ostrom, which is nested organizations. These suborganizations—such as monitoring committees, income-generating cooperatives, and conflict resolution groups—play specific roles within the broader governance framework. This implies distributing responsibilities across multiple levels, articulating CC organizations with international entities, and fostering new productive organizations that improve or reinforce sources of income intertwined with governance systems developed for the REDD+ project. For example, local monitoring groups focus on enforcing sustainable practices, while regional committees coordinate with external stakeholders and manage larger-scale challenges.

As explained to us in some field discussions, REDD+ projects have catalyzed the formation of new organizations or strengthened existing ones, promoting sustainable resource exploitation around fishing, mollusk extraction like piangua (Anadara tuberculosa and A. similis), forest products, and new plantations of coconut and plantain. For instance, the establishment of the Naidí exploitation organization in the Pacific enabled better commercialization strategies and long-term sustainability for this forest product. This organization has integrated several CC that signed separate REDD+ projects, allowing for regional integration. Such nested organizations enhance social capital and governance, aligning individual and collective goals with conservation outcomes. Additionally, this weaving of new networks contributes to strengthening the economic incomes of communities, indirectly compensating opportunity costs and providing greater monetary benefits through improved market access and collective negotiation. This dynamic further explains why, despite not directly covering opportunity costs, these projects can yield significant indirect economic benefits for participants.

Individual and collective benefits. The collective nature of payments in most REDD contracts, in the Pacific region, means that individual opportunity costs for community members involved in deforestation or coca cultivation are not directly compensated. Despite this, a combination of direct and indirect economic benefits, when interacting with collective governance mechanisms, provides a plausible explanation for the observed success.

REDD+ projects generate direct income for community members through resources allocated to producers and former "*corteros* " (loggers). For instance, individuals previously engaged in illicit activities are now trained for roles in forest monitoring, resource management, and sustainable land use practices. This employment transition partly offsets individual opportunity costs but also provides a sense of purpose and involvement in community governance.

Community members highlight the importance of the decision not to distribute monetary income individually to each family. Since the perception of some members of the communities, individual compensation can lead to unintended consequences, such as families using these payments to purchase tools that amplify deforestation activities, like chainsaws, or fishing equipment for unsustainable practices in protected areas. In contrast, collective payments align incentives at the community level, reducing the risk of actions that contradict zero-deforestation objectives.

An important component of the projects supporting REDD+ contracts in the Colombian Pacific has been the development of programs aimed at diversifying agricultural activities and providing sustainable alternatives to coca cultivation. These initiatives include agro-

forestry, sustainable farming, and the management of non-timber forest products, offering communities a variety of options to stabilize their livelihoods. By delivering technical assistance and resources, REDD+ helps ensure stable income streams from legal crops such as cacao, coconut, and plantain. Areas previously dedicated to coca cultivation have transitioned to plantain and coconut farming - as pointed out by community members - illustrating how these projects promote sustainable economic alternatives while reducing dependency on illicit crops.

Collective investment in public goods. A distinctive feature of REDD+ programs in the Colombian Pacific region is their ability to address longstanding deficits in public goods provision. Collective payments from REDD+ contracts are often reinvested in infrastructure projects such as schools, health centers, or communal transportation systems. These investments provide value that exceeds individual compensation for abstaining from illegal activities, demostrating a structural benefit of collective payment systems. However, it is essential to highlight that this characteristic is specific to the REDD+ projects in this region and cannot be generalized to other contexts, such as the Amazon. This localized reinvestment in public goods strengthens collective ties and aligns community-wide benefits with long-term sustainability goals.

In the specific case of coca, REDD+ projects provide additional non-monetary benefits. Abandoning illegal activities reduces legal risks, improves relationships with authorities, and diminishes social stigma within the community. Beyond securing new sources of income, participants gain material benefits, such as access to state support programs, and non-material benefits, including improved social acceptance and community recognition. These benefits create incentives for individuals to transition toward legal livelihoods, enhancing their social standing and expanding opportunities to participate in other governmental or international programs.

Rol of intermediary. Although not a mechanism in itself, the presence of intermediaries, such as FondoAcción, significantly enhances the effectiveness of REDD+ projects. Fondo Acción, which manages eight of the 17 projects analyzed, stands out as an intermediary because it operates as a resource manager for international cooperation funds and does not claim a percentage of profits as seen in other REDD+ contracts. This unique approach contrasts with intermediaries in other contexts and emphasizes the organization's role in ensuring equitable and transparent resource allocation. Additionally, the majority of projects studied under FondoAcción's management include clear action plans that promote sustainable land-use practices and ensure alternative income sources for participating communities. These plans often involve structured support for agricultural diversification and land-use transitions that align with long-term conservation goals. Furthermore, intermediaries bridge cultural and operational gaps between local actors and international stakeholders, ensuring that REDD+ projects are contextually appropriate and effective.

Surveillance and global climate governance. REDD+ projects integrate local governance into broader frameworks of global climate governance through participation in international carbon markets. This engagement introduces external monitoring mechanisms that reinforce behavioral changes. Communities become aware that their environmental performance is continuously observed through satellite imagery and other surveillance tools.

The interaction between external surveillance and local contracts creates a dual accountability system. On one hand, communities are motivated to comply with conservation targets

to meet contractual obligations. On the other, external scrutiny enhances awareness about global environmental responsibilities. This multilayered system of accountability ensures that REDD+ aligns local conservation efforts with international climate goals, positioning these projects as a bridge between local needs and global governance frameworks.

7 Concluding Remarks

Using satellite data and a difference-in-differences design, we document positive, statistically robust impacts after the signing of collective REDD+ agreements. Although illicit coca crops are not the main national driver of deforestation—nor an explicit target of REDD+—their recent expansion into environmentally strategic Afro-Colombian territories poses acute challenges. Our estimates show that REDD+ has contained both deforestation pressures and the spread of coca, preventing additional losses that would otherwise have materialised.

These findings make this study one that adds to the still limited body of causal evidence showing that REDD+ not only works, but works in collective-property settings, where governance dynamics differ sharply from privately owned forests. They also demonstrate that REDD+ can deliver co-benefits that go well beyond carbon, an aspect rarely quantified in the literature.

The results highlight the promise of integrated policies that knit together conservation and drug-policy objectives, while underscoring the critical co-benefits some REDD+ projects can generate. Moreover, the relevance of these outcomes extends beyond regions threatened by coca cultivation, offering valuable insights for communities under high pressure from both legal and illegal land use changes. This is especially true for populations facing persistent poverty, where conservation efforts alone may not directly improve material well-being.

The positive outcomes identified can be attributed to several mechanisms explored in this study: (1) the reinforcement of governance and collective action through strengthened local institutions and social capital, (2) the provision of direct and indirect economic benefits that align community and individual incentives and facilitated the transition of coca cultivators toward sustainable livelihoods, (3) the pivotal role of intermediaries in ensuring equitable and effective implementation, and (4) the integration of local governance into global climate governance frameworks through surveillance and carbon market participation. Together, these mechanisms create a system of accountability and opportunity, enabling REDD+ to address both environmental and social challenges effectively.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT by OpenAI to improve the clarity and readability of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

Acción, F. (2020). Con los pies en el bosque. https://fondoaccion.org/wp-content/uploads/2023/05/Con-los-pies-en-el-bosque-ING.pdf

Agrawal, A. (2007). Forests, Governance, and Sustainability: Common Property Theory and its Contributions. *International Journal of the Commons*, *1*(1). https://doi.org/10.18352/ijc.10

Alix-Garcia, J. M., Shapiro, E. N., & Sims, K. R. E. (2012). Forest Conservation and Slippage: Evidence from Mexico's National Payments for Ecosystem Services Program [Publisher: [Board of Regents of the University of Wisconsin System, University of Wisconsin Press]]. *Land Economics*, *88*(4), 613–638. Retrieved March 29, 2024, from https://www.jstor.org/stable/23272663

Alix-Garcia, J. M., Sims, K. R. E., Orozco-Olvera, V. H., Costica, L. E., Fernández Medina, J. D., & Romo Monroy, S. (2018). Payments for environmental services supported social capital while increasing land management. *Proceedings of the National Academy of Sciences*, *115*(27), 7016–7021. https://doi.org/10.1073/pnas.1720873115

Andersson, K. P., Cook, N. J., Grillos, T., Lopez, M. C., Salk, C. F., Wright, G. D., & Mwangi, E. (2018). Experimental evidence on payments for forest commons conservation [Publisher: Nature Publishing Group]. *Nature Sustainability*, *1*(3), 128–135. https://doi.org/10.1038/s41893-018-0034-z

Angelsen, A. (2008). REDD models and baselines. *International Forestry Review*, *10*(3), 465–475. https://doi.org/10.1505/ifor.10.3.465

Angelsen, A. (2009, January). *Realising REDD+: National Strategy and Policy Options* [Google-Books-ID: vNavVSPOtioC]. CIFOR.

Angelsen, A. (2014). The Economics of REDD+ [Num Pages: 17]. In *Handbook of Forest Resource Economics*. Routledge.

Artaxo, P., Hansson, H. C., Machado, L. A. T., & Rizzo, L. V. (2022). Tropical forests are crucial in regulating the climate on Earth (J. Males, Ed.). *PLOS Climate*, *1*(8), e0000054. https://doi.org/10.1371/journal.pclm.0000054

Badgley, G., Freeman, J., Hamman, J. J., Haya, B., Trugman, A. T., Anderegg, W. R. L., & Cullenward, D. (2022). Systematic over-crediting in California's forest carbon offsets program [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/gcb.15943]. *Global Change Biology*, *28*(4), 1433–1445. https://doi.org/10.1111/gcb.15943

Bayrak, M. M., & Marafa, L. M. (2016). Ten Years of REDD+: A Critical Review of the Impact of REDD+ on Forest-Dependent Communities [Number: 7 Publisher: Multidisciplinary Digital Publishing Institute]. *Sustainability*, *8*(7), 620. https://doi.org/10.3390/su8070620

Bermúdez, A. (2021). El mayor proyecto de bonos de carbono de Colombia podría estar inflando sus cifras de deforestación. *Mongabay Latam*. https://es.mongabay.com/2021/06/bonos-de-carbono-mataven-colombia-redd-investigacion/

Bermúdez, A. (2022a). Sin salvaguardas para los proyectos de carbono, no hay paraíso. *Mongabay Latam*. https://es.mongabay.com/2022/11/sin-salvaguardas-para-los-proyectos-de-carbono-no-hay-paraiso/

Bermúdez, A. (2022b). Un posconflicto marcado por la deforestación: El panorama de los bosques en Colombia. *Mongabay Latam*. https://es.mongabay.com/2022/12/un-posconflicto-marcado-por-la-deforestacion-en-colombia/

Bermúdez, A. (2023a). Colombia: Corte Constitucional examinará por primera vez un conflicto por bonos de carbono [Section: Environmental news]. *Noticias ambientales*. Retrieved April 5, 2024, from https://es.mongabay.com/2023/05/colombia-corte-constitucional-examinara-por-primera-vez-un-conflicto-por-bonos-de-carbono/

Bermúdez, A. (2023b). Colombia: El resguardo indígena que vendió bonos de carbono sin que sus habitantes lo supieran. *Mongabay Latam*. https://es.mongabay.com/2023/06/colombia-el-resguardo-indigena-que-vendio-bonos-de-carbono-sin-que-sus-habitantes-lo-supieran/

Bonilla-Mejía, L., & Higuera-Mendieta, I. (2019). Protected Areas under Weak Institutions: Evidence from Colombia. *World Development*, *122*, 585–596. https://doi.org/10.1016/j.worlddev.2019.06.019

Brienen, R. J. W., Phillips, O. L., Feldpausch, T. R., Gloor, E., Baker, T. R., Lloyd, J., Lopez-Gonzalez, G., Monteagudo-Mendoza, A., Malhi, Y., Lewis, S. L., Vásquez Martinez, R., Alexiades, M., Álvarez Dávila, E., Alvarez-Loayza, P., Andrade, A., Aragão, L. E. O. C., Araujo-Murakami, A., Arets, E. J. M. M., Arroyo, L., … Zagt, R. J. (2015). Long-term decline of the Amazon carbon sink. *Nature*, *519*(7543), 344–348. https://doi.org/10.1038/nature14283

Brombacher, D., Garzón, J. C., & Vélez, M. A. (2021). Introduction special issue: Environmental impacts of illicit economies. *Journal of Illicit Economies and Development*, *3*(1), 1–9.

Brownson, K., Guinessey, E., Carranza, M., Esquivel, M., Hesselbach, H., Madrid Ramirez, L., & Villa, L. (2019). Community-Based Payments for Ecosystem Services (CB-PES): Implications of community involvement for program outcomes. *Ecosystem Services*, *39*, 100974. https://doi.org/10.1016/j.ecoser.2019.100974

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-Differences with multiple time periods. *Journal of Econometrics*, *225*(2), 200–230. https://doi.org/10.1016/j.jeconom.2020.12.001

Camacho, A., & Mejía, D. (2017). The health consequences of aerial spraying illicit crops: The case of colombia. *Journal of Health Economics*, *54*, 147–160.

Castro, C., & WWF. (2023). Trayectorias de la deforestación por consejos comunitarios del pacífico colombiano, 1990-2022.

Charoud, H., Costedoat, S., Izquierdo-Tort, S., Moros, L., Villamayor-Tomás, S., Castillo-Santiago, M. Á., Wunder, S., & Corbera, E. (2023). Sustained participation in a Payments for Ecosystem Services program reduces deforestation in a Mexican agricultural frontier [Publisher: Nature Publishing Group]. *Scientific Reports*, *13*(1), 22314. https://doi.org/10.1038/s41598-023-49725-7

Chervier, C., & Costedoat, S. (2017). Heterogeneous Impact of a Collective Payment for Environmental Services Scheme on Reducing Deforestation in Cambodia. *World Development*, *98*, 148–159. https://doi.org/10.1016/j.worlddev.2017.04.014

Clements, T., John, A., Nielsen, K., An, D., Tan, S., & Milner-Gulland, E. J. (2010). Payments for biodiversity conservation in the context of weak institutions: Comparison

of three programs from Cambodia. *Ecological Economics*, *69*(6), 1283–1291. https://doi.org/10.1016/j.ecolecon.2009.11.010

Correa, J., Cisneros, E., Börner, J., Pfaff, A., Costa, M., & Rajão, R. (2020). Evaluating REDD+ at subnational level: Amazon fund impacts in Alta Floresta, Brazil. *Forest Policy and Economics*, *116*, 102178. https://doi.org/10.1016/j.forpol.2020.102178

Cuenca, P., Robalino, J., Arriagada, R., & Echeverría, C. (2018). Are government incentives effective for avoided deforestation in the tropical Andean forest? [Publisher: Public Library of Science]. *PLOS ONE*, *13*(9), e0203545. https://doi.org/10.1371/journal.pone.0203545

Davalos, L., Bejarano, A., Hall, M., Correa, H., Corthals, A., & Espejo, O. (2011). Forests and Drugs: Coca-Driven Deforestation in Tropical Biodiversity Hotspots. *Environmental science & technology*, *45*. https://doi.org/10.1021/es102373d

Dávalos, L. M. (2021). Los fantasmas del desarrollo pasado. In P. Gootenberg & L. M. Dávalos (Eds.), *Los orígenes de la cocaína. colonización y desarrollo fallido en los andes amazónicos* (pp. 41–86). Universidad de los Andes, Facultad de Economía, Centro de Estudios sobre Seguridad y Drogas.

Dávalos, L. M., Sanchez, K. M., & Armenteras, D. (2016). Deforestation and Coca Cultivation Rooted in Twentieth-Century Development Projects. *BioScience*, *66*(11), 974–982. https://doi.org/10.1093/biosci/biw118

de Chaisemartin, C., & D'Haultfœuille, X. (2020). Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *American Economic Review*, *110*(9), 2964–2996. https://doi.org/10.1257/aer.20181169

Demarchi, G., Subervie, J., Catry, T., & Tritsch, I. (2023). Using publicly available remote sensing products to evaluate REDD+. *Global Environmental Change*, *80*, 102653. https://doi.org/10.1016/j.gloenvcha.2023.102653

DNP. (2023). *Evaluación institucional y de resultados del programa nacional integral de sustitución de cultivos ilícitos (pnis)* (tech. rep.). Unión Temporal Ipsos-Uniandes PNIS 2022. Bogotá.

Engel, S. (2016). The Devil in the Detail: A Practical Guide on Designing Payments for Environmental Services. *International Review of Environmental and Resource Economics*, *9*(1–2), 131–177. https://doi.org/10.1561/101.00000076

Engel, S., Pagiola, S., & Wunder, S. (2008). Designing payments for environmental services in theory and practice: An overview of the issues. *Ecological Economics*, *65*(4), 663–674. https://doi.org/10.1016/j.ecolecon.2008.03.011

Erasso, C., & Vélez, M. A. (2020). *¿los cultivos de coca causan deforestación?* (Tech. rep. No. 5). Documento CESED.

FAO. (2020). *Global Forest Resources Assessment 2020*. https://doi.org/10.4060/ca9825en

FAO & FILAC. (2021). *Los pueblos indígenas y tribales y la gobernanza de los bosques. Una oportunidad para la acción climática en Latina América y el Caribe*. FAO ; retrieved December 28, 2024, from https://openknowledge.fao.org/handle/20.500.14283/cb2953es

Graham, H. A., Cunliffe, A. M., Mitchard, E. T. A., Ruiz Ramos, J., Sáenz, L., Burslem, D. F., & Philipson, C. (2023). Response to west et al. (2023) [Published December 5, 2023, e-Letter/Response]. *Science*, *381*(6660). https://www.science.org/...

Guizar-Coutiño, A., Jones, J. P. G., Balmford, A., Carmenta, R., & Coomes, D. A. (2022). A global evaluation of the effectiveness of voluntary REDD+ projects at reducing deforestation and degradation in the moist tropics. *Conservation Biology*, *36*(6), e13970. https://doi.org/10.1111/cobi.13970

Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S. A., Tyukavina, A., Thau, D., Stehman, S. V., Goetz, S. J., Loveland, T. R., Kommareddy, A., Egorov, A., Chini, L., Justice, C. O., & Townshend, J. R. G. (2013). High-Resolution Global Maps of 21st-Century Forest Cover Change [Publisher: American Association for the Advancement of Science]. *Science*, *342*(6160), 850–853. https://doi.org/10.1126/science.1244693

Haya, B. K., Abayo, A., So, I., & Elias, M. (2023). *Voluntary Registry Offsets Database v10, Berkeley Carbon Trading Project*. University of California, Berkeley.

Hayes, T., Grillos, T., Bremer, L. L., Murtinho, F., & Shapiro, E. (2019). Collective PES: More than the sum of individual incentives. *Environmental Science & Policy*, *102*, 1–8. https://doi.org/10.1016/j.envsci.2019.09.010

Intergovernmental Panel On Climate Change. (2023, July). *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781009157896

Jayachandran, S., de Laat, J., Lambin, E. F., Stanton, C. Y., Audy, R., & Thomas, N. E. (2017). Cash for carbon: A randomized trial of payments for ecosystem services to reduce deforestation [Publisher: American Association for the Advancement of Science]. *Science*, *357*(6348), 267–273. https://doi.org/10.1126/science.aan0568

Lobo, I. D., & Vélez, M. A. (2022). From strong leadership to active community engagement: Effective resistance to illegal coca crops in Afro-Colombian collective territories. *International Journal of Drug Policy*, *102*, 103579. https://doi.org/10.1016/j.drugpo.2022.103579

Londoño, A., Marín, L., & Vélez, M. A. (2024). *Resumen de la evaluación institucional, de resultados y de impacto del pnis: Recomendaciones para un nuevo programa de transformación territorial en zonas cocaleras* (tech. rep.).

Malan, M., Carmenta, R., Gsottbauer, E., Hofman, P., Kontoleon, A., Swinfield, T., & Voors, M. (2024). Evaluating the impacts of a large-scale voluntary REDD+ project in Sierra Leone [Publisher: Nature Publishing Group]. *Nature Sustainability*, *7*(2), 120–129. https://doi.org/10.1038/s41893-023-01256-9

Marín-Llanes, L., Fernández, M., Vélez, M. A., Martínez, E., & Murillo-Sandoval, P. (2024). *Coca-based local growth and its socio-economic impact in colombia* (tech. rep. No. 31). Documento CEDE.

Marketplace, E. (2024). State of the Voluntary Carbon mArket.

Marketplace, F. T. E. (2023). *State of the Voluntary Carbon Markets 2023* (tech. rep.). Forest Trends Association. Washington DC.

Mejía, D., Restrepo, P., & Rozo, S. (2017). On the effects of enforcement on illegal markets: Evidence from a quasi-experiment in colombia. *World Bank Economic Review*, *31*(2), 570–594.

Murillo-Sandoval, P., Van Dexter, K., Van Den Hoek, J., Wrathall, D., & Kennedy, R. (2020). The end of gunpoint conservation: Forest disturbance after the colombian

peace agreement. *Environmental Research Letters*, *15*(3). https://doi.org/10.1088/1748-9326/ab6ae3

Murillo-Sandoval, P. J., Kilbride, J., Tellman, E., Wrathall, D., & Van Den Hoek, J. (2023). The post-conflict expansion of coca farming and illicit cattle ranching in colombia. *Scientific Reports*, *13*(1), 1–12. https://doi.org/10.1038/s41598-023-28918-0

Naime, J., Angelsen, A., Molina-Garzón, A., Carrilho, C. D., Selviana, V., Demarchi, G., Duchelle, A. E., & Martius, C. (2022). Enforcement and inequality in collective PES to reduce tropical deforestation: Effectiveness, efficiency and equity implications. *Global Environmental Change*, *74*, 102520. https://doi.org/10.1016/j.gloenvcha.2022.102520

Oliveira Fiorini, A. C., Mullally, C., Swisher, M., & Putz, F. E. (2020). Forest cover effects of payments for ecosystem services: Evidence from an impact evaluation in Brazil. *Ecological Economics*, *169*, 106522. https://doi.org/10.1016/j.ecolecon.2019.106522

Ostrom, E. (1990). The Rudiments of a Theory of the Origins, Survival, and Performance of Common Property Institutions.

Peña, X., Vélez, M. A., Cárdenas, J. C., Perdomo, N., & Matajira, C. (2017). Collective Property Leads to Household Investments: Lessons From Land Titling in Afro-Colombian Communities [Publisher: Elsevier Ltd]. *World Development*, *97*, 27–48. https://doi.org/10.1016/j.worlddev.2017.03.025

Pfaff, A., Rodriguez, L. A., & Shapiro-Garza, E. (2019). Collective Local Payments for ecosystem services: New local PES between groups, sanctions, and prior watershed trust in Mexico. *Water Resources and Economics*, *28*, 100136. https://doi.org/10.1016/j.wre.2019.01.002

Probst, B., Toetzke, M., Kontoleon, A., Diaz Anadon, L., & Hoffmann, V. H. (2023). *Systematic review of the actual emissions reductions of carbon offset projects across all major sectors* (Working Paper) (Accepted: 2023-07-06T13:37:00Z). ETH Zurich. https://doi.org/10.3929/ethz-b-000620307

Quiroga, V., Stevenson, P., & Wagner, H. (2022). The effect of illicit crops on forest cover in colombia. *Journal of Land Use Science*, *17*(1), 47–59.

Rincón-Ruiz, A., Pascual, U., & Flantua, S. (2013). Examining spatially varying relationships between coca crops and associated factors in Colombia, using geographically weight regression. *Applied Geography*, *37*, 23–33. https://doi.org/10.1016/j.apgeog.2012.10.009

Robalino, J., & Pfaff, A. (2013). Ecopayments and Deforestation in Costa Rica: A Nationwide Analysis of PSA's Initial Years [Publisher: [Board of Regents of the University of Wisconsin System, University of Wisconsin Press]]. *Land Economics*, *89*(3), 432–448. Retrieved April 6, 2024, from https://www.jstor.org/stable/24243718

Romero, M., & Saavedra, S. (2020). Communal Property Rights and Deforestation [Publisher: Routledge]. *The Journal of Development Studies*. Retrieved November 29, 2020, from https://www.tandfonline.com/doi/abs/10.1080/00220388.2020.1817394

Roopsind, A., Sohngen, B., & Brandt, J. (2019). Evidence that a national REDD+ program reduces tree cover loss and carbon emissions in a high forest cover, low deforestation country [Publisher: Proceedings of the National Academy of Sciences]. *Proceedings of the National Academy of Sciences*, *116*(49), 24492–24499. https://doi.org/10.1073/pnas.1904027116

Roth, J., Sant'Anna, P. H. C., Bilinski, A., & Poe, J. (2023). What's trending in difference-in-differences? A synthesis of the recent econometrics literature. *Journal of Econometrics*, *235*(2), 2218–2244. https://doi.org/10.1016/j.jeconom.2023.03.008

Rubiano Lizarazo, M., Vélez Lesmes, M. A., Restrepo Díaz, D., & Ramos Torres, B. (2020). *¿reanudar la fumigación aérea con glifosato de cultivos ilícitos en colombia? un resumen de la literatura científica* (tech. rep.). Universidad de los Andes. Bogotá.

Schmid, D., & Castro, C. (2023). *REDD+ projects in the Colombian Amazon: Social challenges and lack of transparency* (tech. rep.). Rainforest Foundation Norway. Oslo.

Schmid, D., & Castro, C. (2025). *New scenarios of non-violent conflict in indigenous communities: The case of redd+ in the colombian amazon* [Forthcoming at Environment and Security Journal].

Simonet, G., Subervie, J., Ezzine-de-Blas, D., Cromberg, M., & Duchelle, A. E. (2019). Effectiveness of a REDD+ Project in Reducing Deforestation in the Brazilian Amazon [_eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1093/ajae/aay028]. *American Journal of Agricultural Economics*, *101*(1), 211–229. https://doi.org/10.1093/ajae/aay028

SINCHI. (2023). Diagnóstico de proyectos REDD+ en la Amazonia colombiana. https://www.sinchi.org.co/diagnostico-de-proyectos-redd-en-la-amazonia-colombiana

Sun, L., & Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics*, *225*(2), 175–199. https://doi.org/10.1016/j.jeconom.2020.09.006

Tracewski, Ł., Butchart, S. H. M., Donald, P. F., Evans, M., Fishpool, L. D. C., & Buchanan, G. M. (2016). Patterns of twenty-first century forest loss across a global network of important sites for biodiversity. *Remote Sensing in Ecology and Conservation*, *2*(1), 37–44. https://doi.org/10.1002/rse2.13

UNDP. (2023). Panorama de la pobreza en el sector rural — Programa De Las Naciones Unidas Para El Desarrollo. Retrieved April 18, 2023, from https://www.undp.org/es/colombia/speeches/panorama-pobreza-sector-rural

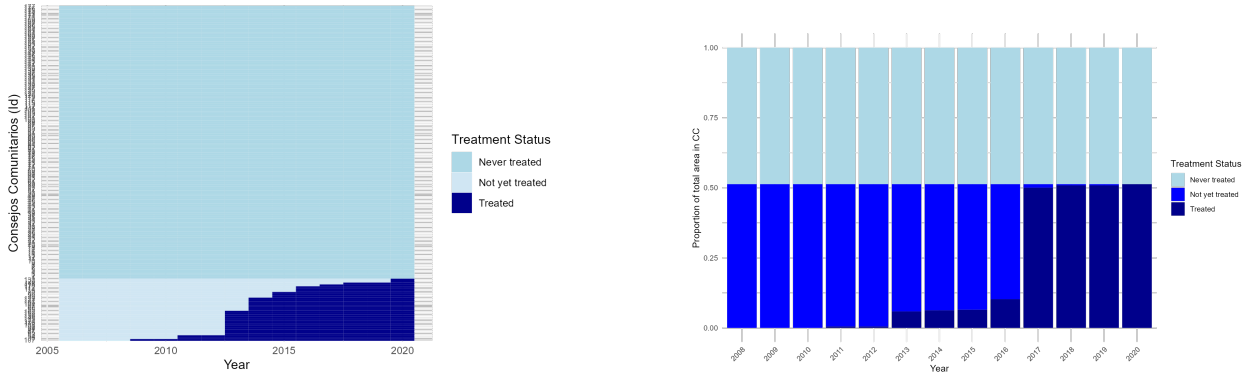
United Nations Office on Drugs and Crime. (2023, October). *Practical Guide: Alternative Development and the Environment*. United Nations. https://doi.org/10.18356/9789213585672

USAID. (2022). BIOREDD+. Programa Páramos y bosques. https://pdf.usaid.gov/pdf_docs/PA00ZRFT.pdf

Vélez, M. A., Robalino, J., Cardenas, J. C., Paz, A., & Pacay, E. (2020). Is collective titling enough to protect forests? Evidence from Afro-descendant communities in the Colombian Pacific region [Publisher: Elsevier Ltd]. *World Development*, *128*. https://doi.org/10.1016/j.worlddev.2019.104837

Wegner, G. I. (2016). Payments for ecosystem services (PES): A flexible, participatory, and integrated approach for improved conservation and equity outcomes. *Environment, Development and Sustainability*, *18*(3), 617–644. https://doi.org/10.1007/s10668-015-9673-7

West, T. A. P., Börner, J., Sills, E. O., & Kontoleon, A. (2020). Overstated carbon emission reductions from voluntary REDD+ projects in the Brazilian Amazon [Publisher: Proceedings of the National Academy of Sciences]. *Proceedings of the National Academy of Sciences*, *117*(39), 24188–24194. https://doi.org/10.1073/pnas.2004334117



(a) Annual progression of REDD+ treatments over time.

(b) Proportional area treated over time.

Figure A.1: Progression and proportional area of REDD+ treatments over time.

Notes: Panel (a) shows the annual progression of REDD+ contracts signed over time. Panel (b) illustrates the proportion of total treated area by year, highlighting the spatial adoption of REDD+ programs.

West, T. A. P., Wunder, S., Sills, E. O., Börner, J., Rifai, S. W., Neidermeier, A. N., Frey, G. P., & Kontoleon, A. (2023). Action needed to make carbon offsets from forest conservation work for climate change mitigation [Publisher: American Association for the Advancement of Science]. *Science*, *381*(6660), 873–877. https://doi.org/10.1126/science.ade3535

Wiik, E., d'Annunzio, R., Pynegar, E., Crespo, D., Asquith, N., & Jones, J. P. G. (2019). Experimental evaluation of the impact of a payment for environmental services program on deforestation [_eprint: https://conbio.onlinelibrary.wiley.com/doi/pdf/10.1111/csp2.8]. *Conservation Science and Practice*, *1*(2), e8. https://doi.org/10.1111/csp2.8

Wilebore, B., Voors, M., Bulte, E. H., Coomes, D., & Kontoleon, A. (2019). Unconditional Transfers and Tropical Forest Conservation: Evidence from a Randomized Control Trial in Sierra Leone. *American Journal of Agricultural Economics*, *101*(3), 894–918. https://doi.org/10.1093/ajae/aay105

Wooldridge, J. M. (2021). Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3906345

Appendix

Descriptive data

The annual progression of treatments highlights the growth in participation over time, with a sharp increase in the number of treated areas starting around 2013. This pattern aligns with the global momentum of REDD+ initiatives, emphasizing their adoption in the study region.

Table A.1: REDD+ Projects in Community Councils (CC)

Project	**Sign Year**	**First Verification**	**Duration**	**Inversor**
The Chocó-Darién Conservation	2009	2012	30 years	Protect / ecoPar
Delfines Cupica REDD+	2011	2019	30 years	Biofix
Concosta REDD+ Project	2013	2017	30 years	Fondo Acción
Sivirú, Usaragá, Pizarro y Pilizá (SUPP)	2013	2017	30 years	Fondo Acción
Río Pepé-ACABA REDD+ Project	2013	2017	30 years	Fondo Acción
(ACAPA-BMF)	2013	2017	30 years	Fondo Acción
Cajambre REDD+ Project	2013	2017	30 years	Fondo Acción
Bajo Calima y Bahía Málaga	2013	2017	30 years	Fondo Acción
Riscales REDD+ Project	2014	2019	30 years	Fondo Acción
Carmen del Darién (CDD)	2014	2017	30 years	Fondo Acción
Proyecto Arlequín REDD+	2015	2021	30 years	Biofix
NAYA REDD+	2016	2022	30 years	Biofix
COCOMAN Frontera	2016	2020	30 years	Biofix
COCOMACIA Community	2017	2021	30 years	Terra
Project Chonta Corozo REDD+	2018	2021	20 years	Biofix
Condoto REDD+	2019	2021	30 years	Terra / Carbo
Pedeguita Jiguamiandó REDD+	2020	2021	40 years	Terra / Carbo

Table A.2: Summary Statistics: Accumulated Deforestation

Cohort	**Obs**	**Mean**	**SD**	**Min**	**Max**
Never	62,535,840	44.85	481.86	0.00	15333.76
2009	247,335	49.80	440.61	0.00	7974.33
2011	2,090,235	11.75	223.24	0.00	10926.75
2013	11,541,915	17.35	259.48	0.00	11239.36
2014	3,317,580	36.91	436.54	0.00	9387.63
2015	1,524,915	60.29	517.98	0.00	7974.33
2016	6,030,900	25.47	319.99	0.00	12065.60
2017	13,161,675	16.64	270.42	0.00	13705.54
2018	1,759,950	2.97	103.99	0.00	7932.06
2019	1,693,845	38.37	401.07	0.00	7974.33
2020	1,953,255	169.28	919.79	0.00	13960.26

Note: This table presents descriptive statistics for accumulated deforestation by cohort. Resolution: 90x90 meter grid cell. Data are based on deforestation measurements across different years.

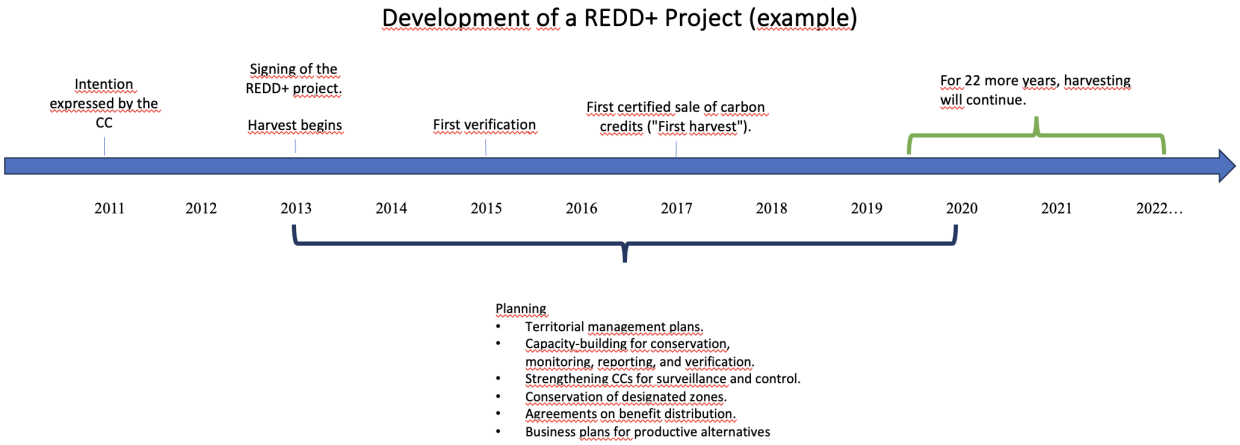


Figure A.2: REDD+ timeline example.

Notes: Note: As an example we use the ACAPA–Bajo Mira y Frontera (ACAPA-BMF) REDD+ project. After the agreement was signed on 2013, the first two years were devoted to drafting the Project and 30-year implementation plan, updating community by-laws and benefit-sharing rules, mapping conservation, production and traditional-use zones, and training governance bodies and patrol units. Between years 2 and 7 the project finalised forest- and land-use plans, set up plans for cocoa, coconut and açaí value chains, installed nurseries and basic processing infrastructure, and launched priority social investments such as potable-water and health facilities. The first third-party verification was scheduled for January 2016, with annual audits thereafter to enable the issuance of VCUs. From year 8 to year 30 the focus shifts to consolidating productive activities, maintaining patrolling and monitoring.

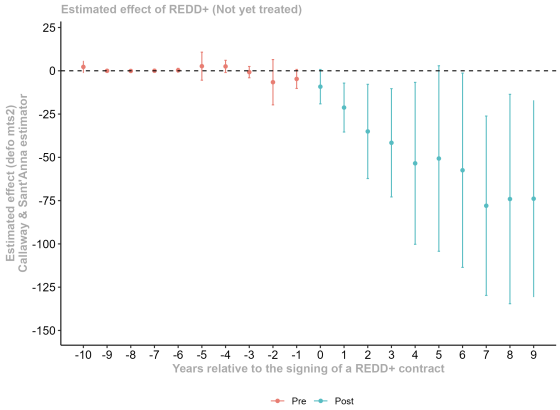
Table A.3: Summary Statistics Coca Crops

Cohort	Obs	Mean	SD	Min	Max
Never	558,263	0.54	2.50	0.00	72.61
2009	2,210	0.00	0.04	0.00	1.50
2011	18,972	0.00	0.02	0.00	1.80
2013	105,791	0.17	0.84	0.00	30.32
2014	30,311	0.03	0.44	0.00	27.24
2015	13,940	0.01	0.11	0.00	4.95
2016	55,284	0.29	1.67	0.00	42.55
2017	89,675	0.00	0.05	0.00	3.64
2018	16,116	0.06	0.43	0.00	9.18
2019	15,572	0.03	0.31	0.00	11.09
2020	17,782	0.10	0.81	0.00	26.60

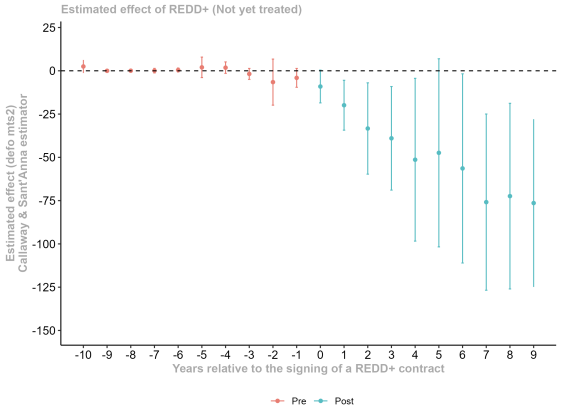
Note: This table presents descriptive statistics for coca crops by cohort. Mean indicates the average area of coca crops in each grid. Data are presented in hectares per square kilometer.

Robustness Checks: Deforestation

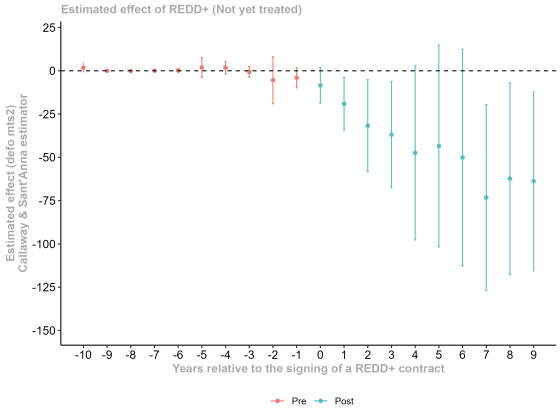
Alternative databases



(a) Sample 2 (5%)



(b) Sample 3 (5%)



(c) Sample 4 (10%)

Figure A.3: Deforestation event study by sample size (5% and 10%). Each plot shows the estimated impact of REDD+ on deforestation.

Notes: The figure shows coefficients and 95% confidence intervals from event study using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. The first two plots are based on 5% stratified random samples of the total observations, each using a different seed. The third plot is based on a stratified random sample of 10% of the total observations. All samples are stratified at the Community Council level. The unit of observation is a pixel-year.

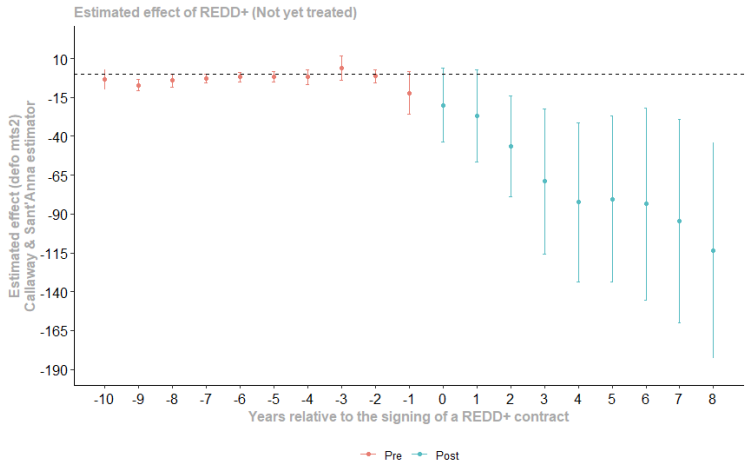
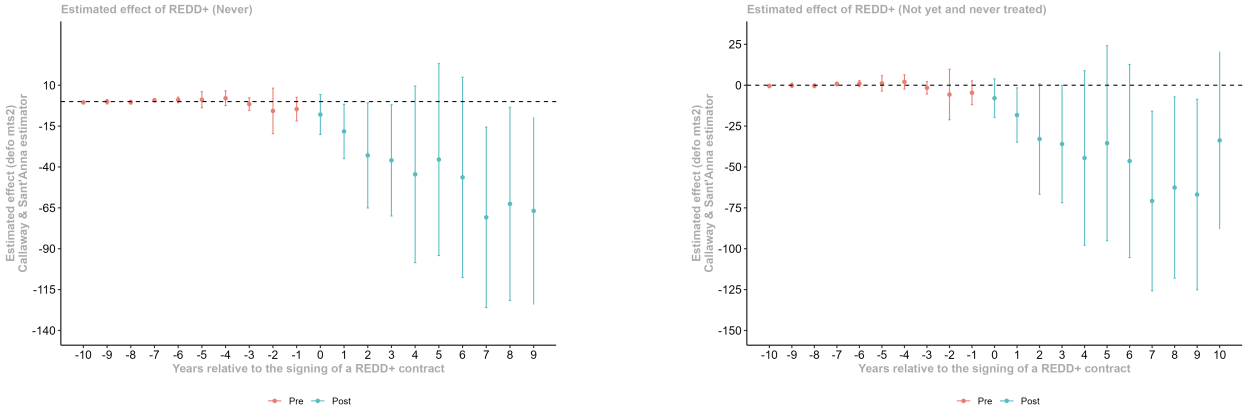


Figure A.4: Event Study: Impact of REDD+ on Deforestation (C&S 2021). Alternative database: Hansen, 2022

Notes: The figure shows coefficients and 95% confidence intervals from event study regressions using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Results are based on a stratified random sample of 5% of the total observations in Hansen database, stratified at the Community Council level. The unit of observation is pixel-year.

Control groups



(a) Control group (Never treated)

(b) Control group (Not yet and never)

Figure A.5: Impact of REDD+ on deforestation by different groups of control.

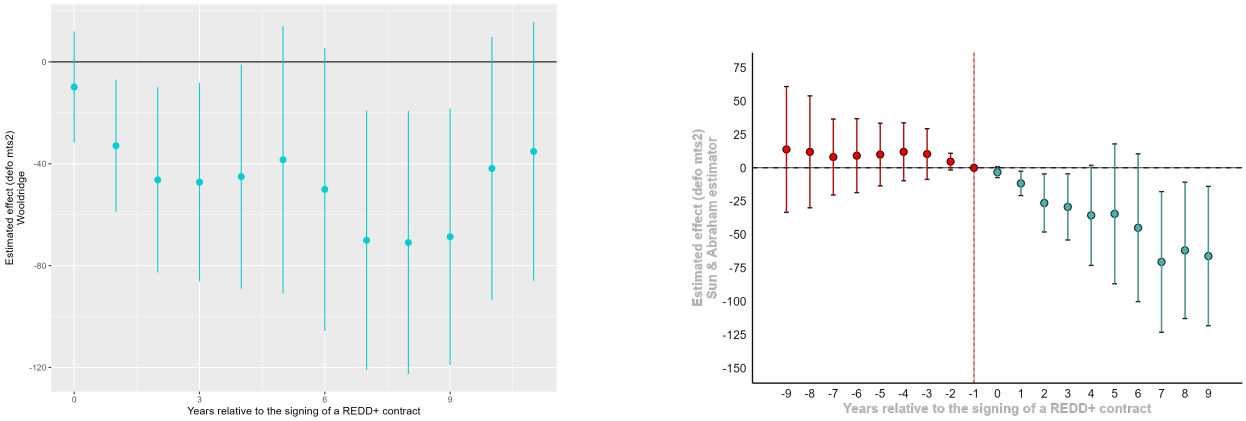
Notes: The figure shows coefficients and 95% confidence intervals from event study using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Table A.5: Impact of REDD+ on Deforestation by different groups of control

	C&S	C&S
ATT	-32.054*	-32.088*
Std. Error	14.39	14.50
Control Group	Never Treated	Not Yet and Never Treated

Notes: This table presents the results of difference-in-difference staggered regressions estimating the average impact of REDD+ signature on deforestation using two control groups: "Never Treated" and "Not Yet and Never Treated. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. Regressions include CC and year fixed effects. No additional covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC (Sample 1). The unit of observation is pixel-year. The estimation was done in the R `did`. Estimation is based on the doubly robust method Callaway and Sant'Anna (2021).

Alternative regression

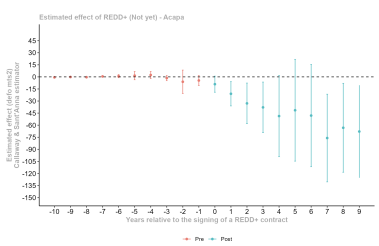


(a) *Wooldridge, 2021* (b) *Sun & Abraham 2021*

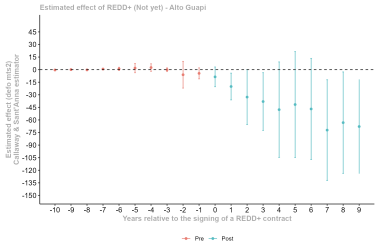
Figure A.6: Alternative estimators.

Notes: Each plot shows coefficients and 95% confidence intervals from event study regressions using Wooldridge and Sun Abraham estimators. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included, and the control group comprises units never treated in the corresponding year. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

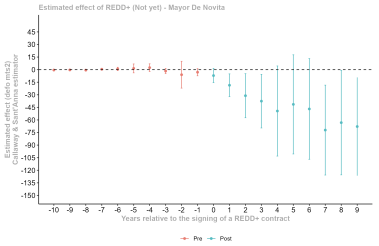
Excluding a CC



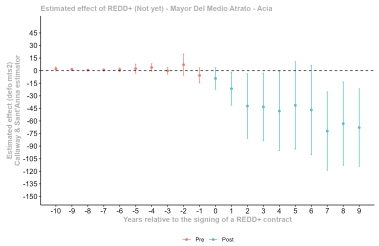
(a) *Except: Acapa*



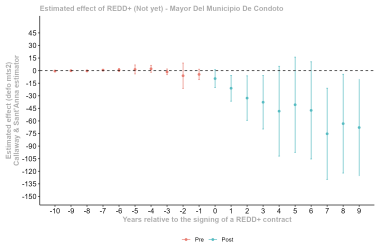
(b) *Except: Alto Guapi*



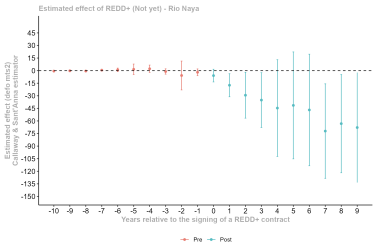
(c) *Except: Mayor de Nóvita*



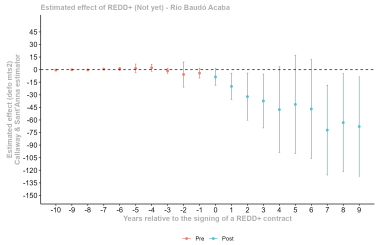
(d) *Except: Mayor del Medio Atrato*



(e) *Except: Mayor del municipio de Condoto*



(f) *Except: Río Naya*



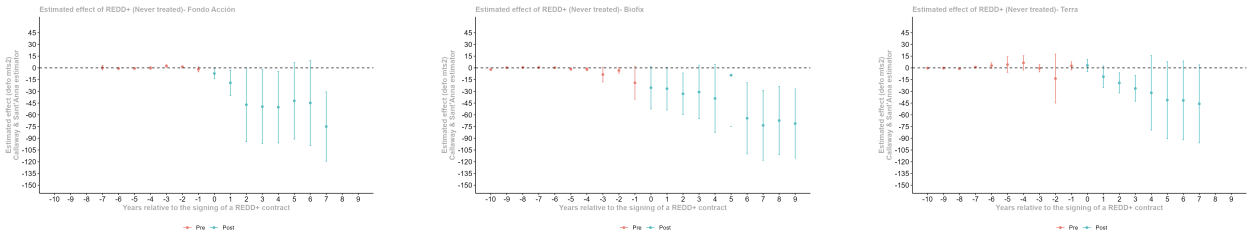
(g) *Except: Río Baudó Acaba*

Figure A.7: Robustness Test: Excluding One Treated Unit.

Notes: Each plot shows the estimated impact of REDD+ on coca when excluding one treated unit. The proportion of area of each Community Council (CC) in the total of CCs of the Pacific is: (a) 1.5%, (b) 1.7%, (c) 2.1%, (d) 9.7%, (e) 1.6%, (f) 3.3%, (g) 3.2%. The seven selected CCs correspond to those treated with a proportion of area greater than 1.5% of the total. It shows coefficients and 95% confidence intervals from event study using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included. Standard errors are clustered at the CC level. The unit of observation is pixel-year.

Heterogeneous Effects: Deforestation

Intermediary

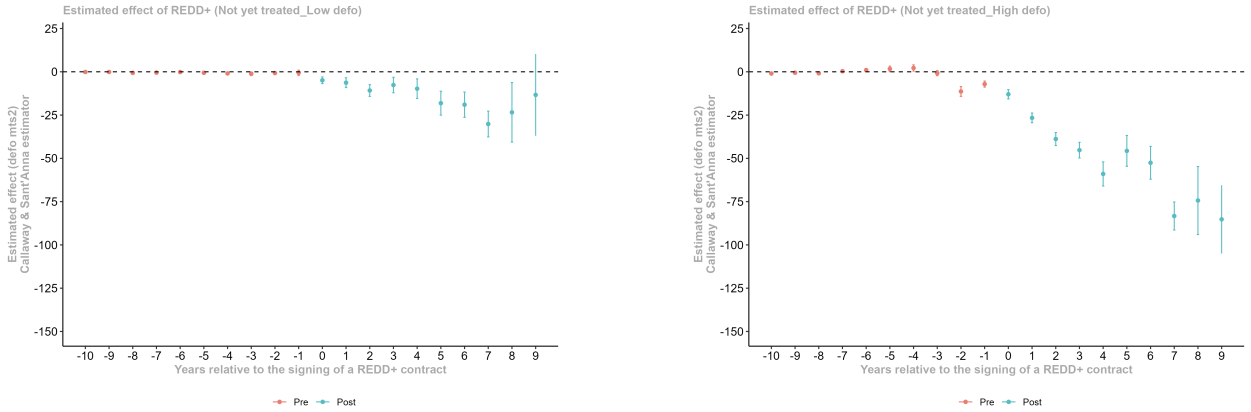


(a) *Fondo Acción* (b) *Biofix* (c) *Terra*

Figure A.8: Heterogeneous effects by intermediary (deforestation).

Notes: The plot shows the estimated impact of REDD+ on deforestation for different project intermediaries. Figure shows coefficients and 95% confidence intervals from event study using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Deforestation pre-trends

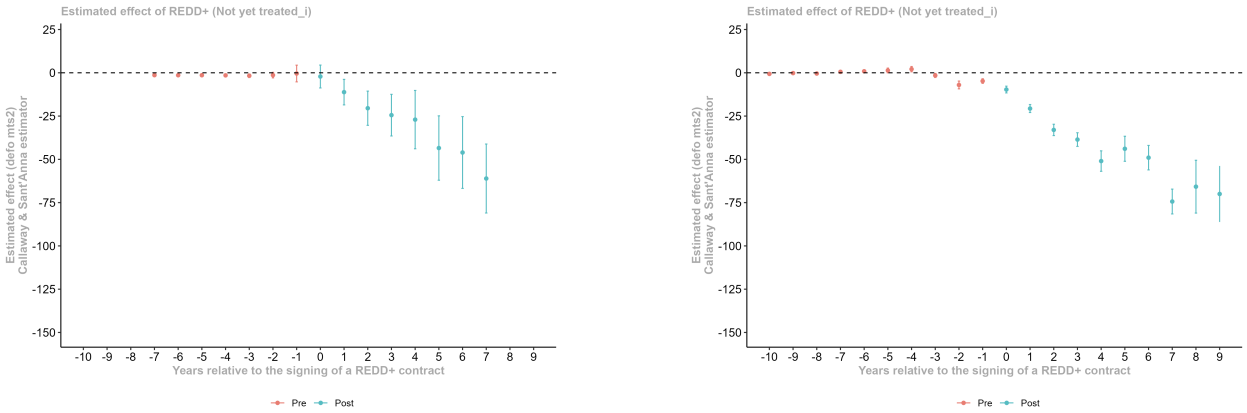


(a) *Low deforestation pre-treatment* (b) *High deforestation pre-treatment*

Figure A.9: Heterogeneous effects by pre-treatment deforestation levels.

Notes: The sample was divided based on median pre-treatment deforestation levels. The left plot shows results for areas with low pre-treatment deforestation, while the right plot shows results for areas with high pre-treatment deforestation. Each plot presents an event study regression analyzing the impact of REDD+ on cumulative deforestation extent. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Community area



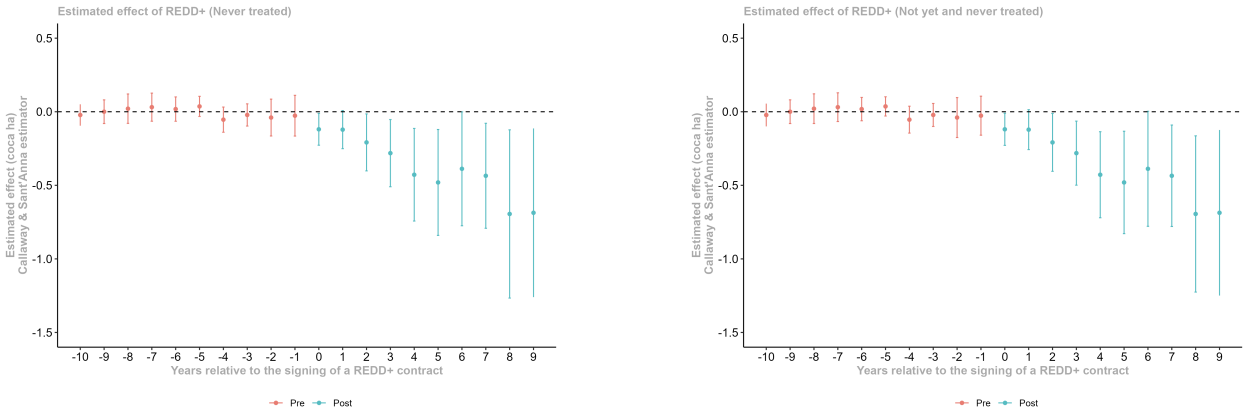
(a) *Small community size* (b) *Large community size*

Figure A.10: Heterogeneous effects by community council size.

Notes: The sample was divided based on median community council area. The left plot shows results for smaller CCs, while the right plot shows results for larger CCs. Each plot shows coefficients and 95% confidence intervals from event study regressions using the Callaway and Sant'Anna (2021) estimator. Standard errors are clustered at the CC level. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. No covariates were included. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Robustness checks: Coca

Control groups



(a) Control group (Never treated)

(b) Control group (Not yet and never)

Figure A.11: Impact of REDD+ on Coca Crops by different groups of control.

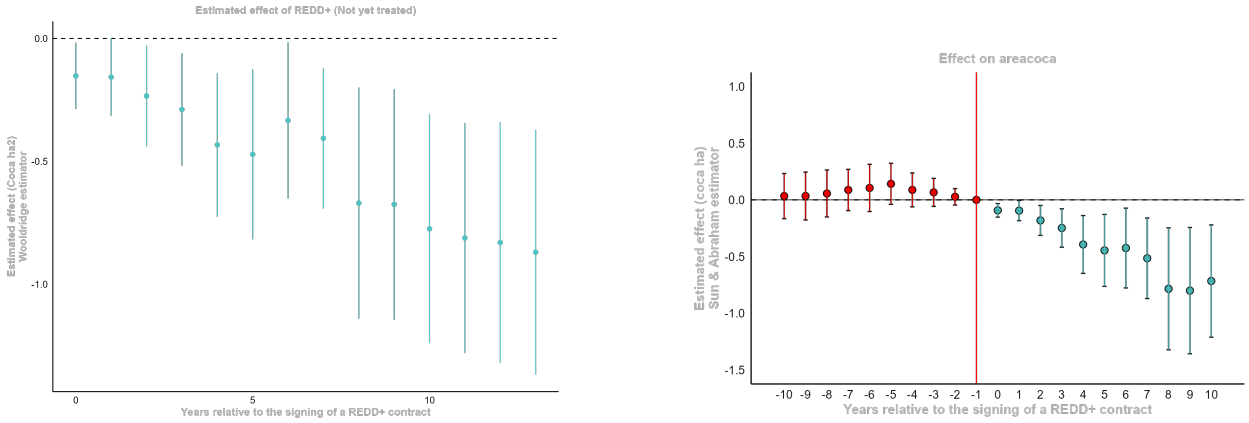
Notes: Each plot shows coefficients and 95% confidence intervals from event study regressions using the Callaway and Sant'Anna (2021) estimator. The outcome variable is coca in ha at 1km2 pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Table A.6: Impact of REDD+ on Coca Cultivation by Different Groups of Control

	C&S	C&S
ATT	-0.337*	-0.337*
Std. Error	0.104	0.106
Control Group	Never Treated	Not Yet and Never Treated

Notes: This table presents the results of difference-in-difference staggered regressions estimating the average impact of REDD+ signature on deforestation using two control groups: "Never Treated" and "Not Yet and Never Treated. The outcome variable is deforestation in square meters at 90x90 meter pixel resolution. Regressions include CC and year fixed effects. No additional covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC (Sample 1). The unit of observation is pixel-year. The estimation was done in the R `did`. Estimation is based on the doubly robust method CS(2021).

Alternative regression

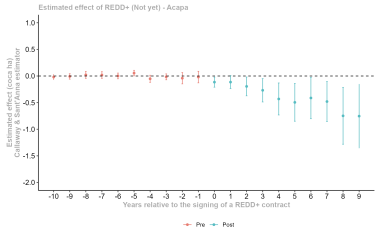


(a) *Wooldridge, 2021* (b) *Sun & Abraham 2021*

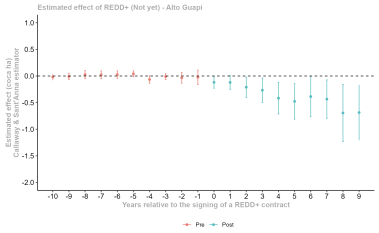
Figure A.12: Alternative estimators.

Notes: Each plot shows coefficients and 95% confidence intervals from event study regressions using the Wooldridge (2021) and Sun & Abraham (2021) estimators. The outcome variable is coca in ha at 1km2 pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

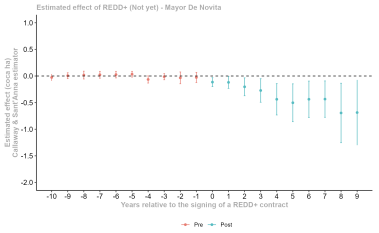
Excluding a CC



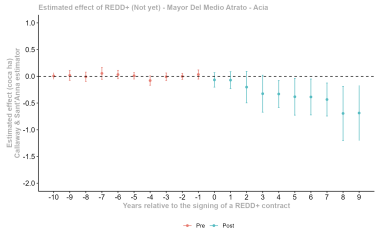
(a) *Except: Acapa*



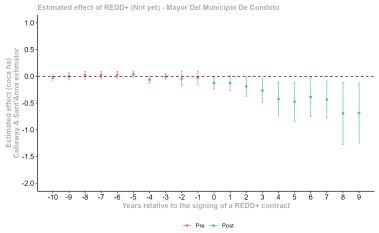
(b) *Except: Alto Guapi*



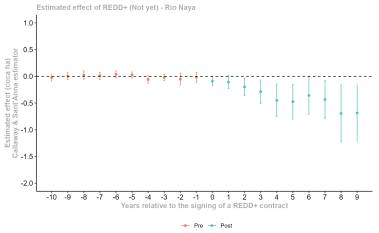
(c) *Except: Mayor de Nóvita*



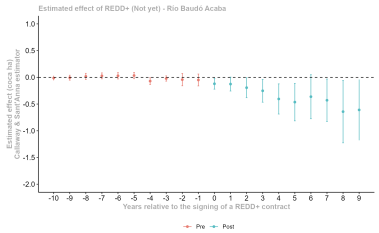
(d) *Except: Mayor del Medio Atrato*



(e) *Except: Mayor del municipio de Condoto*



(f) *Except: Río Naya*

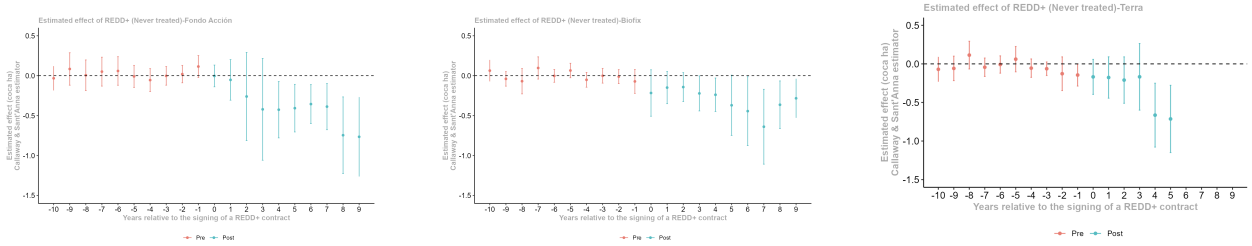


(g) *Except: Río Baudó Acaba*

Figure A.13: Robustness Test: Excluding One Treated Unit.

Notes: Each plot shows the estimated impact of REDD+ on coca when excluding one treated unit. The proportion of area of each CC in the total of CCs of the Pacific is: (a) 1.5%, (b) 1.7%, (c) 2.1%, (d) 9.7%, (e) 1.6%, (f) 3.3%, (g) 3.2%. The seven selected CCs correspond to those treated with a proportion of area greater than 1.5% of the total. Each plot shows coefficients and 95% confidence intervals from event study using the Callaway and Sant'Anna (2021) estimator. The outcome variable is coca in ha at 1km2 pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Heterogeneous Effects: Coca Intermediary

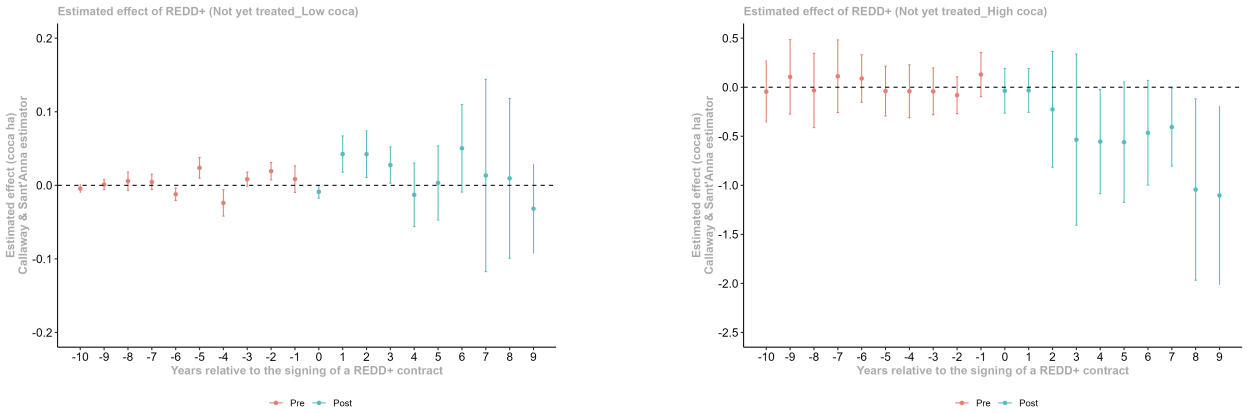


(a) *Fondo Acción* (b) *Biofix* (c) *Terra*

Figure A.14: Heterogeneous effects by intermediary (Coca).

Notes: Each plot shows the estimated impact of REDD+ on deforestation for different project intermediaries. Each plot presents an event study regression analyzing the impact of REDD+ on coca crop extent. Each plot shows coefficients and 95% confidence intervals from event study regressions using the Callaway and Sant'Anna (2021) estimator. The outcome variable is coca in ha at 1km2 pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Coca pre-trends

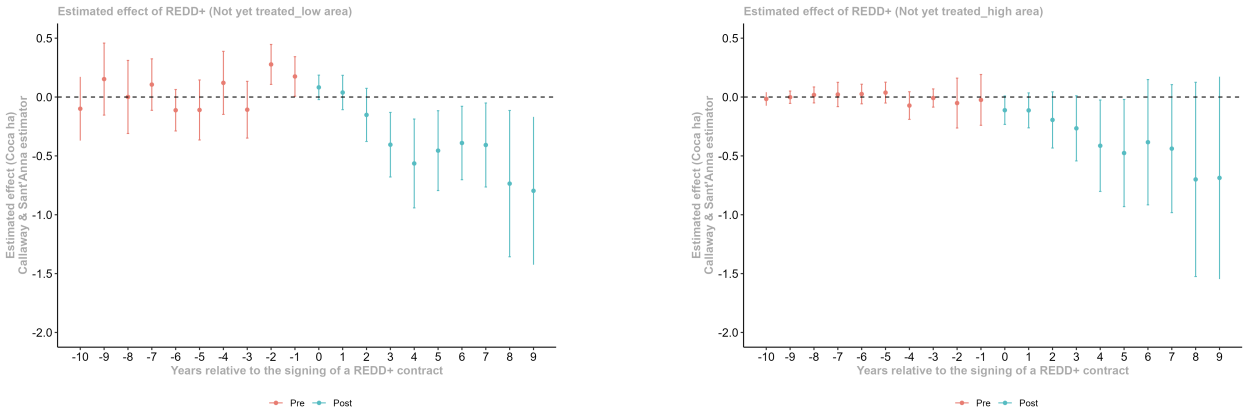


(a) *Low coca pre-treatment* (b) *High coca pre-treatment*

Figure A.15: Heterogeneous effects by pre-treatment coca levels.

Notes: The sample was divided based on median pre-treatment coca crop presence. The left plot shows results for areas with low pre-treatment coca presence, while the right plot shows results for areas with high pre-treatment coca presence. Each plot presents an event study regression analyzing the impact of REDD+ on coca crop extent. Each plot shows coefficients and 95% confidence intervals from event study regressions using the Callaway and Sant'Anna (2021) estimator. The outcome variable is coca in ha at 1km2 pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.

Community area



(a) *Small community size* (b) *Large community size*

Figure A.16: Heterogeneous effects by community council size (coca).

Notes: The sample was divided based on median community council area. The left plot shows results for smaller CCs, while the right plot shows results for larger CCs. Each plot presents an event study regression analyzing the impact of REDD+ on coca crop extent. Each plot shows coefficients and 95% confidence intervals from event study regressions using the Callaway and Sant'Anna (2021) estimator. The outcome variable is coca in ha at 1km2 pixel resolution. No covariates were included, and the control group comprises units not yet treated in the corresponding year. Standard errors are clustered at the CC level. Results are based on a stratified random sample of 5% of the total observations, stratified at the CC. The unit of observation is pixel-year.